



CAPSTONE PROJECT REPORT

Report 6 – Software User Guides

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I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

II. Release Package & User Guides

1. Deliverable Package

No.	Deliverable Item	Description
1	Project Schedule/Tracking	The project timeline showing all phases, iterations, milestones, and deadlines, with tasks assigned to each member and their actual progress tracked against the plan.
2	Project Backlog	The prioritized list of all epics, user stories, and use cases of the system, including status, effort estimation, and the sprint in which each item is delivered.
3	Source Codes	The complete source code of the system: the Next.js web client, the four NestJS microservices (Gateway, IAM, Clinical EMR, Payment), the Python AI services, and all Docker and configuration files needed to build and run the system.
4	Database Script(s)	The database initialization script that creates one logical database per service, together with the TypeORM migration scripts that build the schema and the seed scripts that populate demo data.
5	Final Report Document	The full capstone report covering the introduction, software requirement specification, system design, implementation, installation guide, testing results, and conclusion.
6	Test Cases Document	The test documentation listing unit test cases and use-case test cases with their input data, expected results, actual results, and pass/fail status, together with the overall test report.
7	Slide	The presentation slides used for the final defense, summarizing the problem, objectives, system architecture, key features, demo scenario, and results achieved.

2. Installation Guides

2.1 System Requirements

For Server:

- **Operating system:** Ubuntu 20.04+ / Windows 10+ (with WSL 2)
- **CPU:** Intel Core i5 (10th Gen or newer) or AMD Ryzen 5 3600 or higher (4 cores)
- **RAM:** 8 GB
- **Storage:** SSD 256 GB

For Client:

- **Operating system:** Ubuntu 20.04+ / Windows 10+ (with WSL 2)
- **Browser:** Google Chrome, Safari, Edge, Firefox, Brave, Opera, etc

2.2 Installation Instruction

2.2.1 Install Docker Desktop

Step 1 — Enable WSL 2 (Windows only)

Windows Subsystem for Linux (WSL) 2 is a prerequisite for Docker Desktop on Windows; it provides the lightweight Linux kernel that Docker containers run on.

1. Open PowerShell as Administrator.

2. Run:

`wsl --install`

3. Restart the computer if prompted.

Step 2 — Install Docker Desktop

1. Download [Docker Desktop](#) for your operating system from the official Docker website

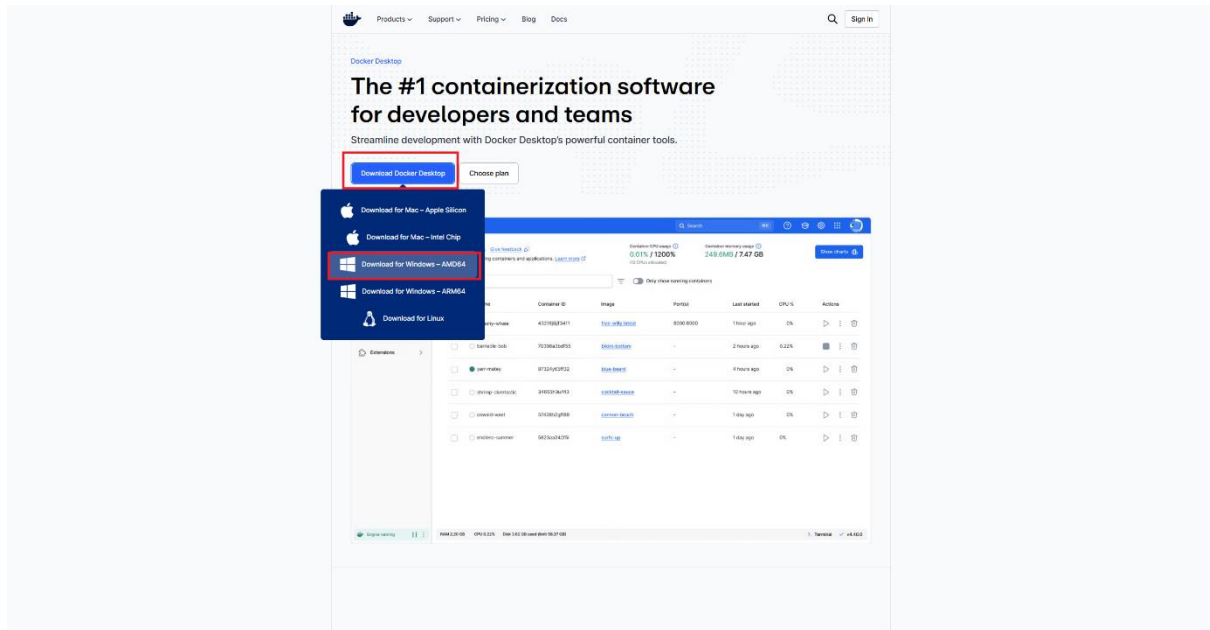


Figure 1. Download Docker Destop

2. Run the installer and follow the on-screen instructions, keeping the "Use WSL 2 instead of Hyper-V" option enabled on Windows.

3. Launch Docker Desktop after installation. You may be prompted to log out and back in.

Step 3 — Configure and verify

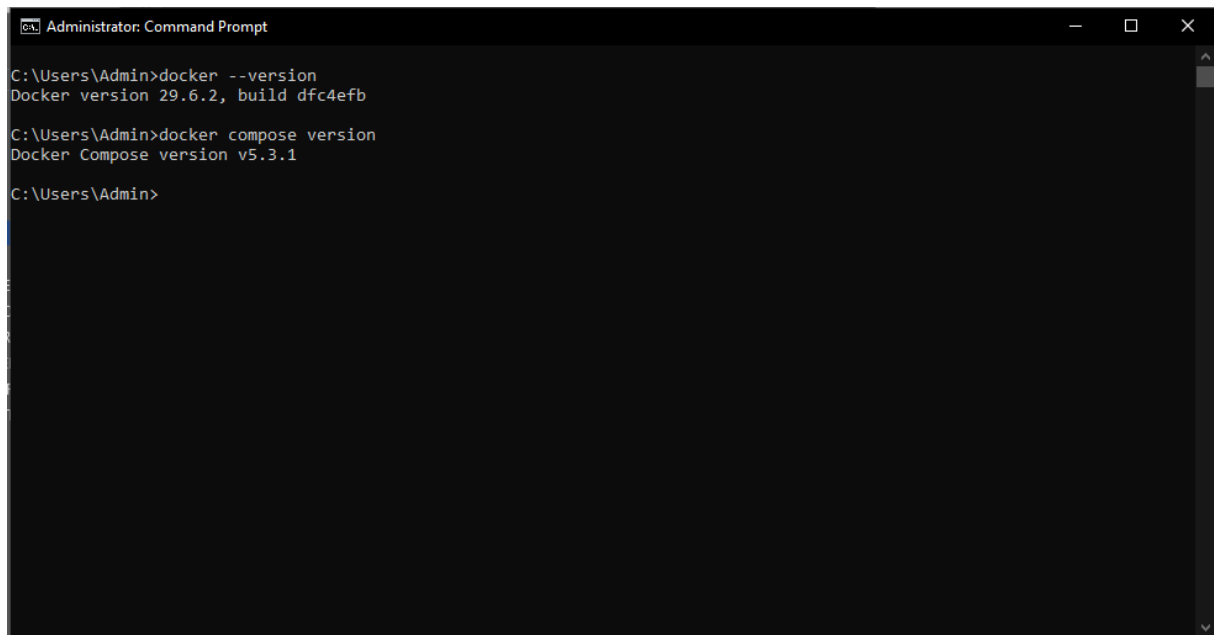
1. Open Settings from the Docker icon in the system tray.

2. Under Resources, allocate at least 4 CPUs and 8 GB of memory — the stack runs six application containers plus Postgres and Redis. Default settings are not sufficient on Windows/macOS.

3. Verify the installation from a terminal:

`docker --version`

`docker compose version`



```
Administrator: Command Prompt

C:\Users\Admin>docker --version
Docker version 29.6.2, build dfc4efb

C:\Users\Admin>docker compose version
Docker Compose version v5.3.1

C:\Users\Admin>
```

Figure 2. Setup Docker Desktop

2.2.2 Install Node.js 20 LTS

Step 1 — Download Installer

Go to the official download page (<https://nodejs.org/en/download>).

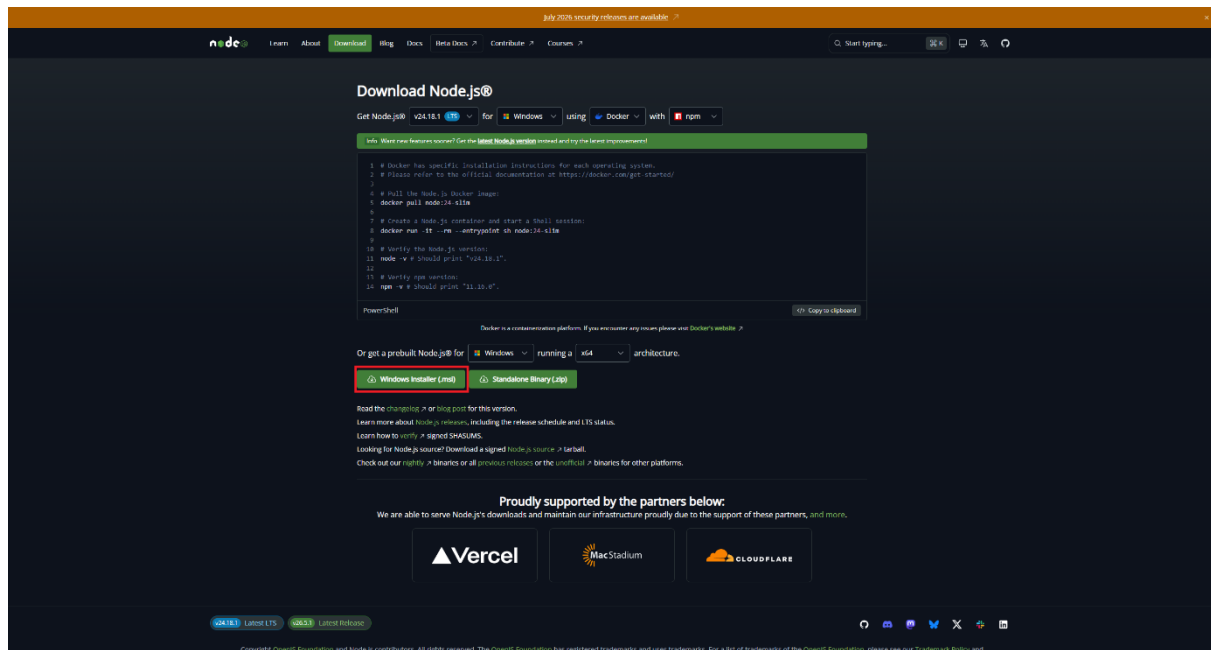


Figure 3. Download NodeJS

Download the Node.js 20 LTS installer (select the 64-bit package matching your operating system).

Step 2 — Execute Setup

Run the downloaded installer and accept the default settings.

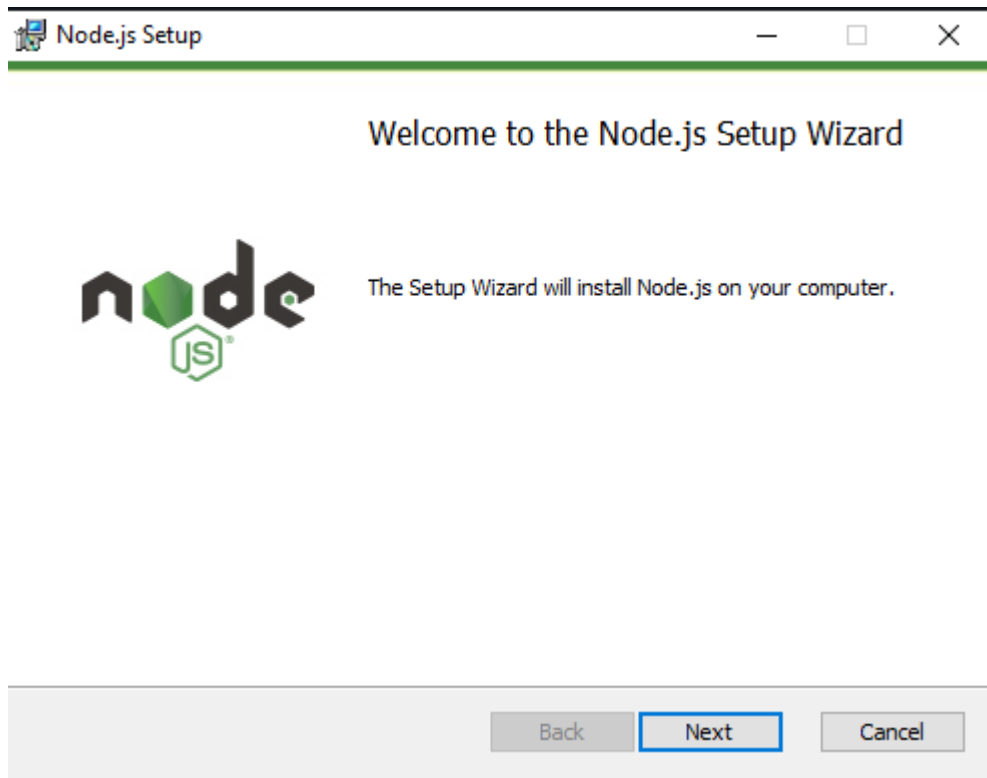


Figure 4. Setup NodeJS

Ensure the Add to PATH option remains checked so node and npm commands can be accessed from any terminal.

Step 3 — Verify Installation

Open a terminal and run the following commands to confirm:

Bash

`node --version` # Expected output: v20.x.x

`npm --version` # Expected output: 10.x.x


```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Admin>node --version
v22.18.0

C:\Users\Admin>npm --version
10.9.3

C:\Users\Admin>
```

Figure 5. Setup NodeJS

2.2.3 Installing Python 3.11+

Step 1 — Download Installers

Navigate to Python Downloads (<https://www.python.org/downloads/>).



Step 2 — Execute Installation

Launch the installer for your OS.

Crucial: Check the box Add python.exe to PATH on the first screen before clicking Install Now.

Step 3 — Set Up Local Virtual Environments

Service 1: KYC OCR Service (kyc_ocr_service)

Note: Heavy ML dependencies (PyTorch, TorchVision, Ultralytics, ONNX Runtime). Initial setup will download several gigabytes; ensure a stable network connection.

Verify Python installation:

Bash

```
python3.11 --version # Expected output: 3.11.x
```

Navigate to the service directory, create, and activate the virtual environment:

Bash

```
cd ai/kyc_ocr_service
```

```
python3.11 -m venv .venv
```

Activate virtual environment

```
source .venv/bin/activate # On macOS/Linux
```

```
# .venv\Scripts\activate # On Windows (PowerShell/CMD)
```

Install dependencies:

Bash

```
pip install -r requirements.txt
```

Service 2: Booking LangGraph Service (booking_langgraph_service)

Verify Python installation:

Bash

```
python3.13 --version # Expected output: 3.13.x
```

Navigate to the service directory, create, and activate the virtual environment:

Bash

```
cd ai/booking_langgraph_service
```

```
python3.13 -m venv .venv
```

Activate virtual environment

source .venv/bin/activate # On macOS/Linux

.venv\Scripts\activate # On Windows (PowerShell/CMD)

Install dependencies:

Bash

pip install -r requirements.txt

2.2.4 Installing and Configuring PostgreSQL 16

Step 1 — Download Installer

Go to the official download page (<https://www.postgresql.org/download>) and choose your operating system.

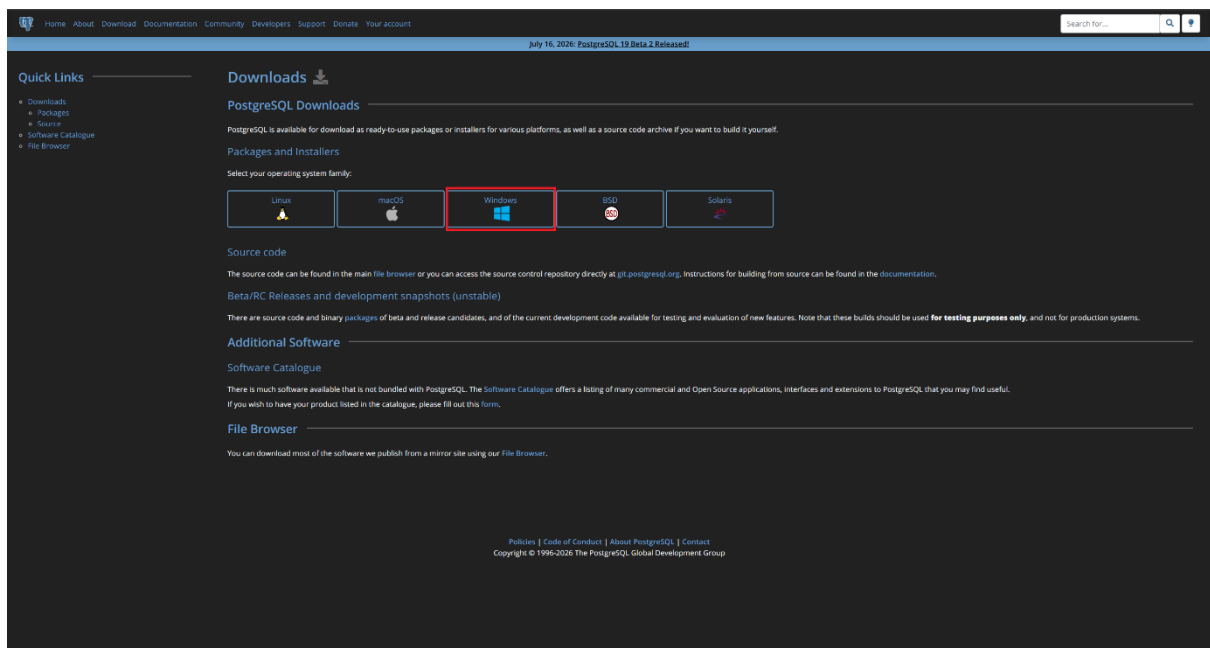


Figure 6. Install Postgresql

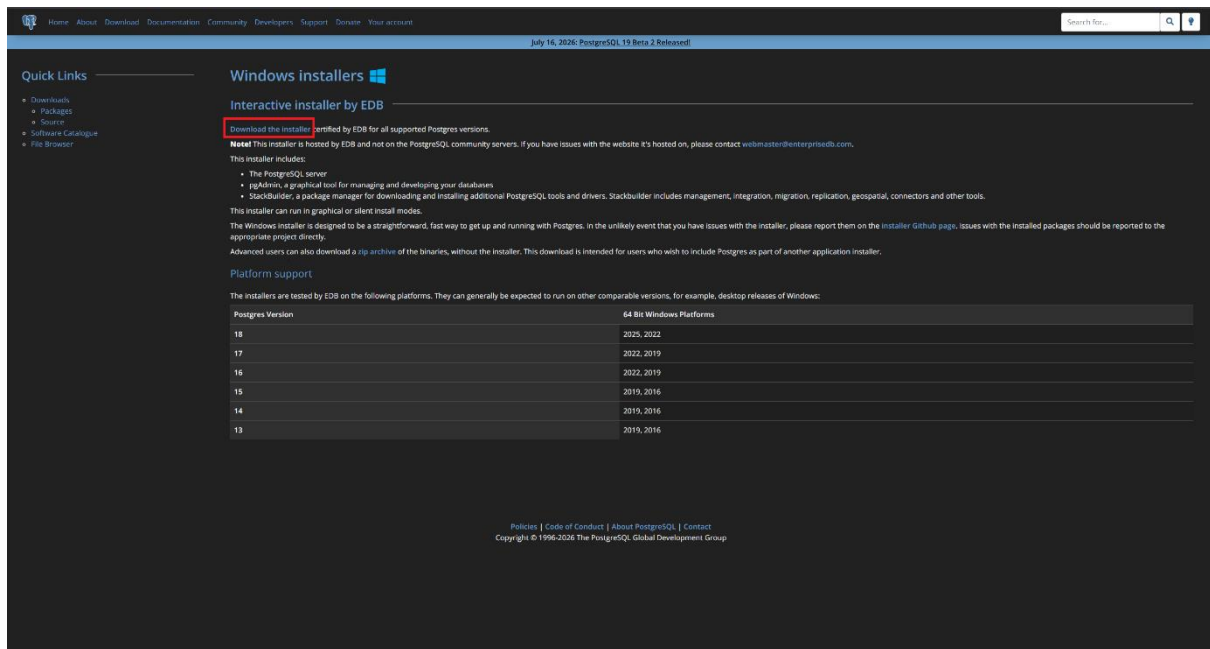


Figure 7. Install Postgresql (2)

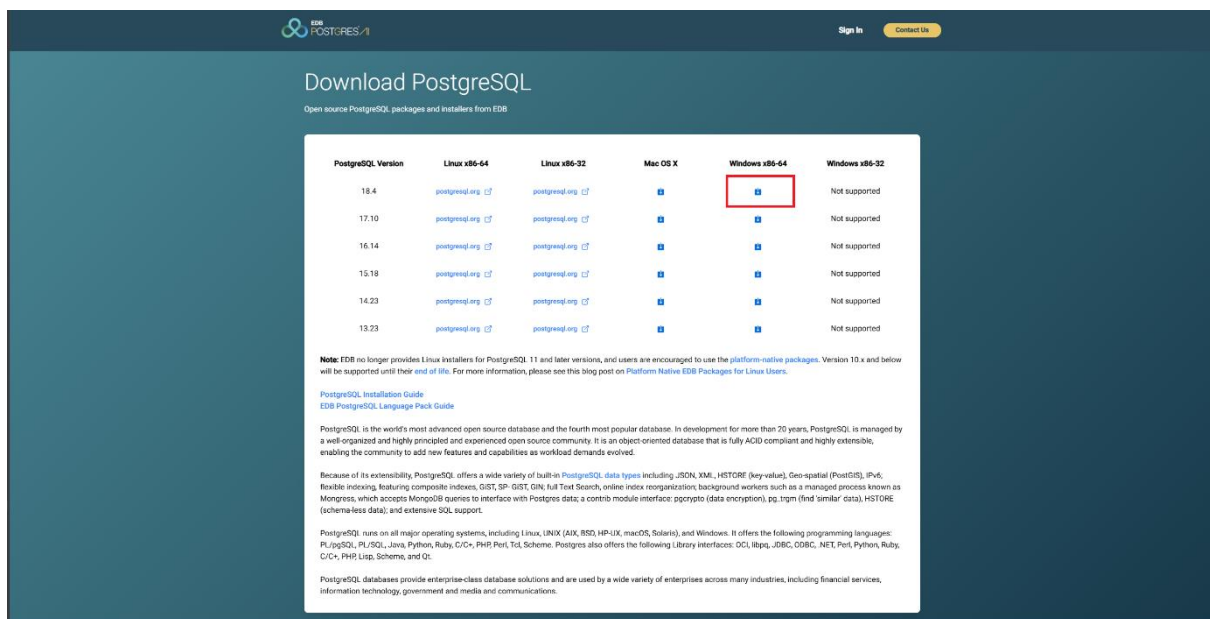


Figure 8. Install Postgresql (3)

On Windows, download the EnterpriseDB installer for **PostgreSQL 16** (64-bit).

Step 2 — Installation Process

Windows Setup

1. Run the installer and accept the default installation directory and data directory.

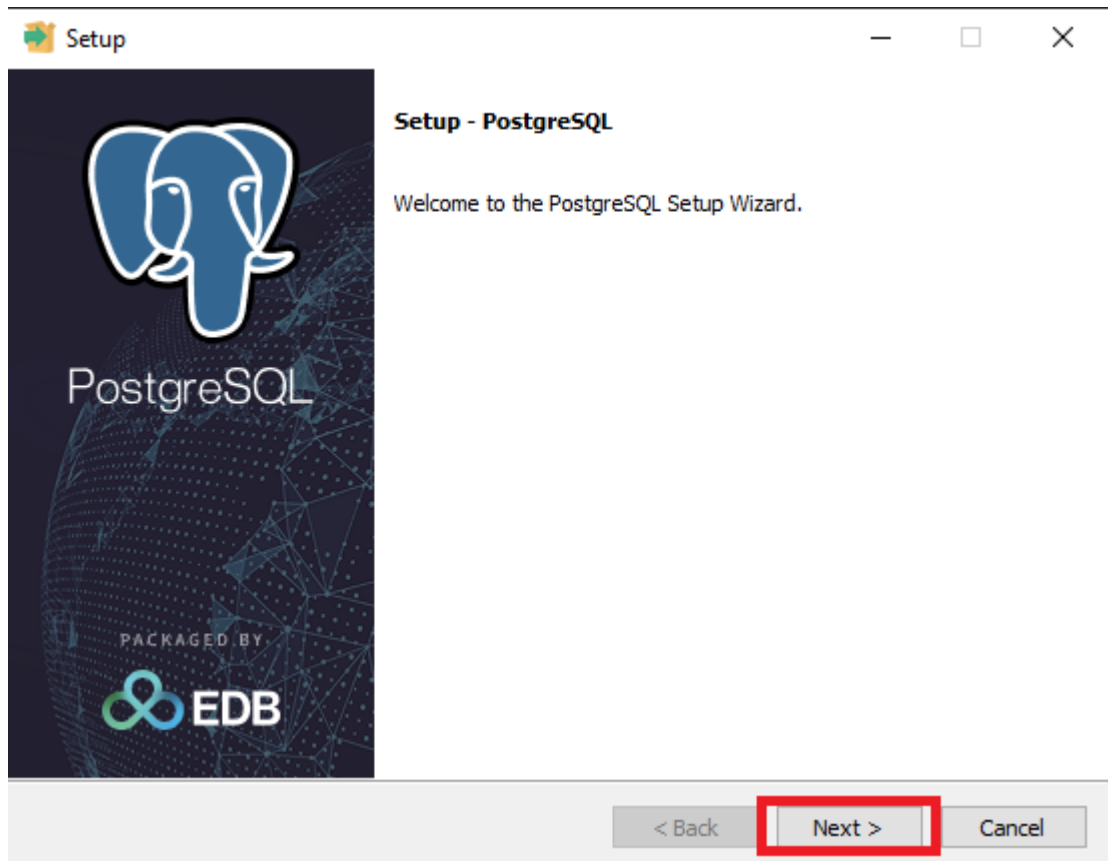


Figure 9. Setup Postgresql

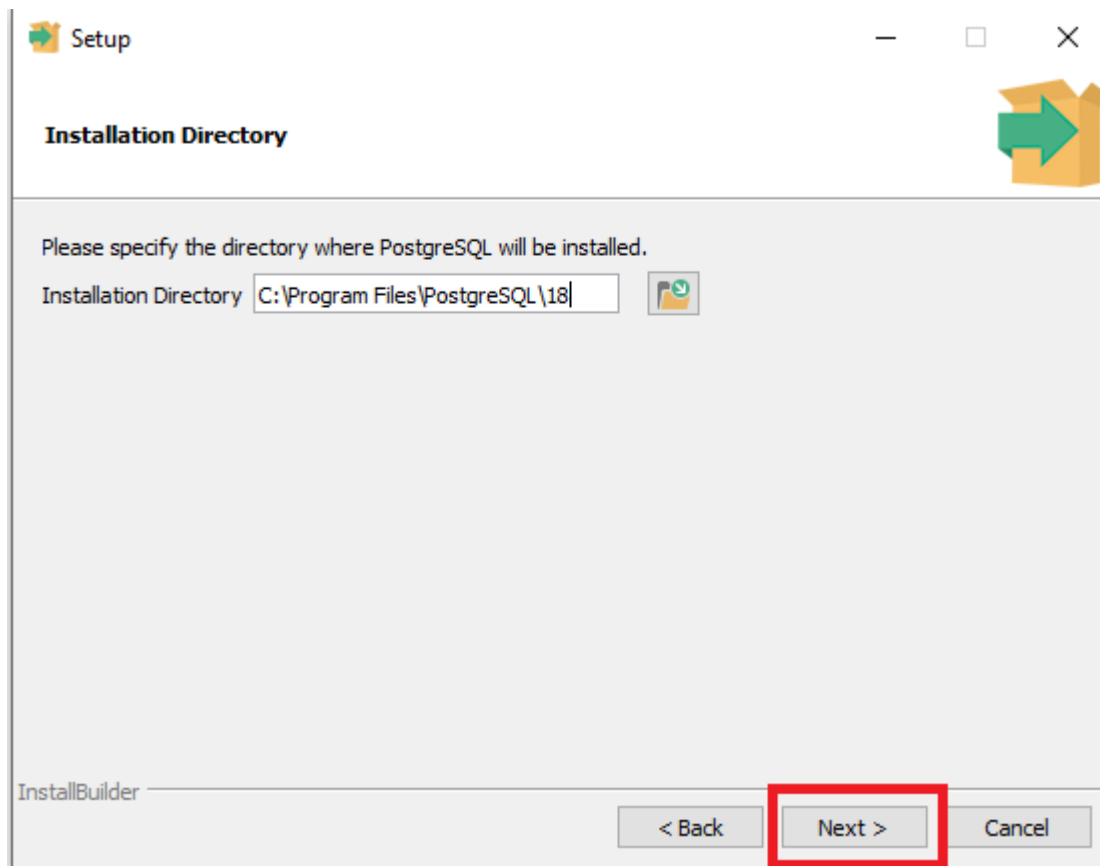


Figure 10. Setup Postgresql (2)

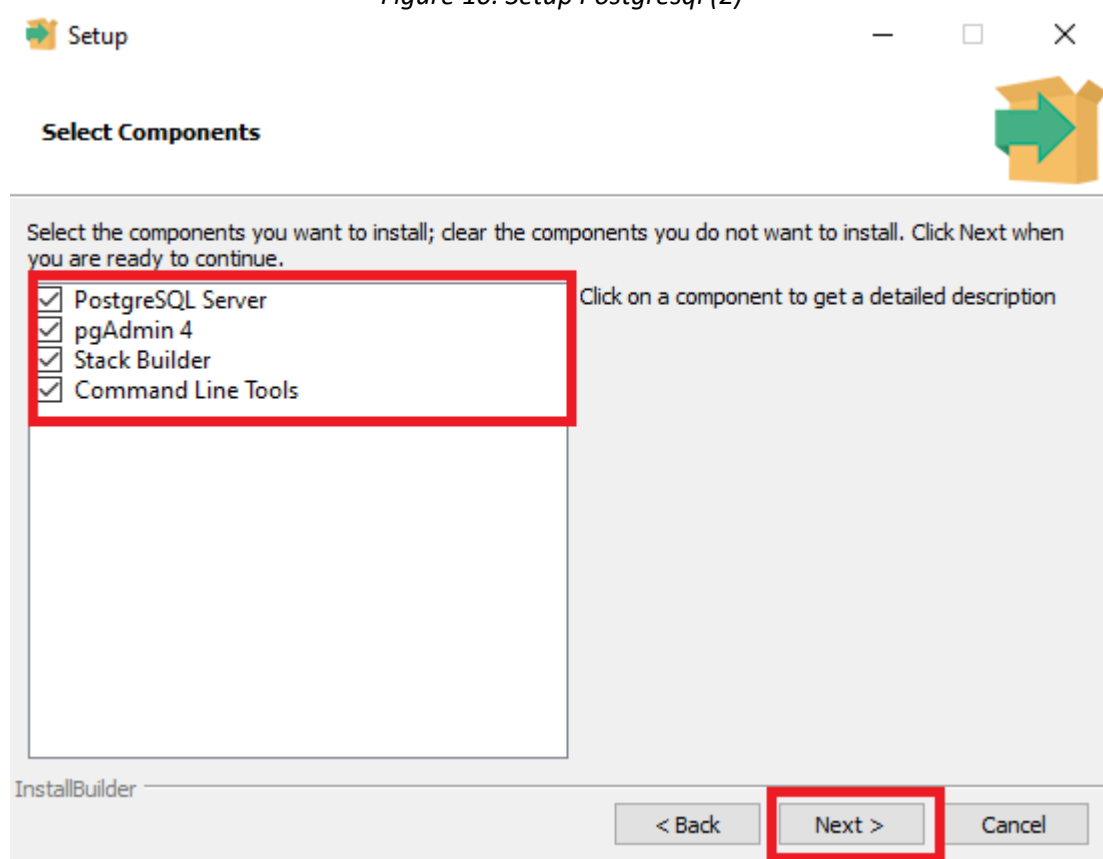


Figure 11. Setup Postgresql (3)

2. Ensure **PostgreSQL Server**, **pgAdmin 4**, and **Command Line Tools** remain selected.
3. Set a password for the postgres superuser (ensure it matches POSTGRES_PASSWORD defined in your .env file).
4. Keep the default port **5432** and default locale, then complete the wizard.

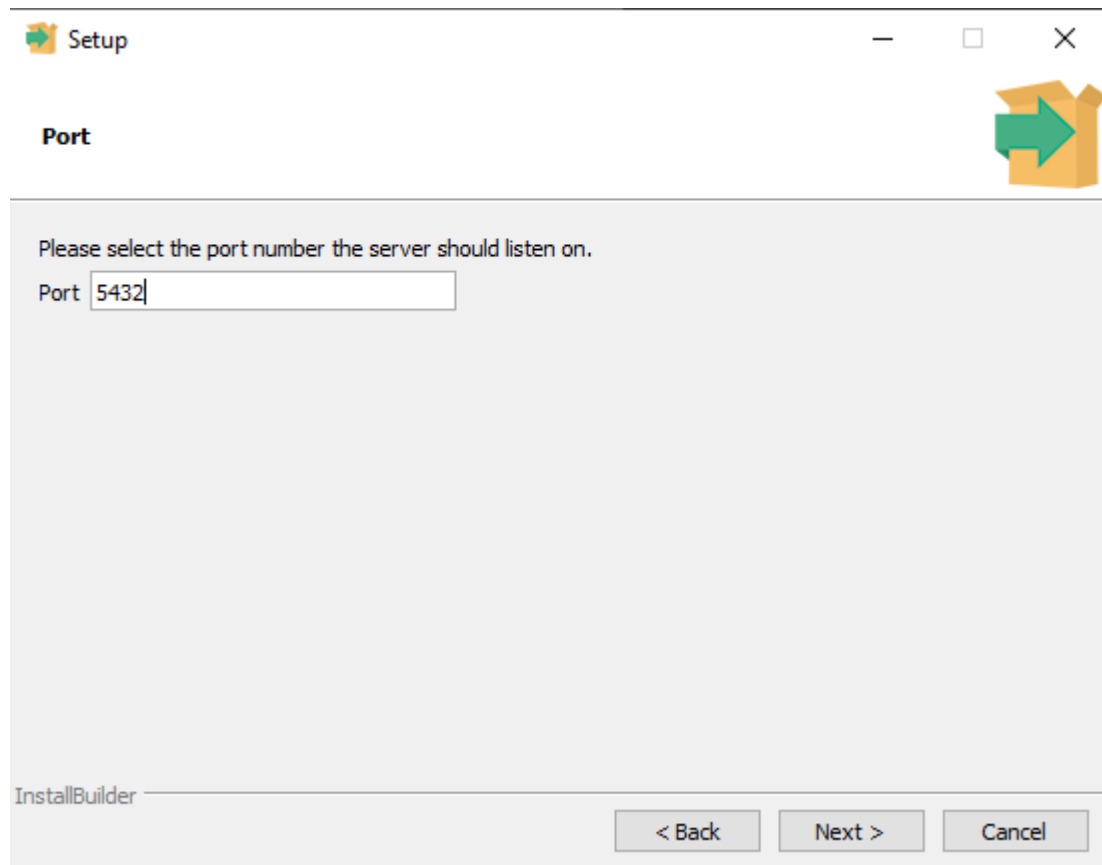


Figure 12. Setup Postgresql (4)

Step 3 — Verify Connection

Confirm that the client binary and connection setup are working:

Bash

```
psql --version    # Expected output: psql (PostgreSQL) 16.x
psql -h localhost -U postgres -c "SELECT version();" 
```

Step 4 — Initialize Logical Databases

Each microservice uses its own dedicated logical database. If running PostgreSQL natively instead of via Docker Compose, execute the initialization script manually:

Bash

```
psql -h localhost -U postgres -f docker/init-db.sql
```

Databases Created:

- auth_service_db & account_service_db (IAM Service)
- core_medical_service_db & core_clinic_service_db (Clinical EMR Service)
- payment_service_db (Payment Service)
- booking_orchestrator_db (AI Booking Service)

Step 5 — Configure Host Environment Files

Update the database connection host inside each microservice's .env file from the Docker service container name to localhost:

```
DB_HOST=localhost
DB_PORT=5432
DB_USERNAME=postgres
DB_PASSWORD=postgres
```

3.1 Overview

This user manual presents the principal operational workflows of the S.M.I.L.E dental platform: staff identity verification, appointment booking, clinical examination, payment and refund handling, and administrative oversight. Each workflow is documented through the English user interface with a concise action sequence and supporting screenshots.

3.2 FLOW 1 — STAFF IDENTITY VERIFICATION

3.2.1 Purpose and Outcome

This flow verifies a staff identity document through OCR and administrator review. The completed submission reached VERIFIED status in the English interface.

3.2.2 Step-by-Step Execution Guide

[Actor: STAFF]

Step 1: Open the login page and authenticate with a staff account.

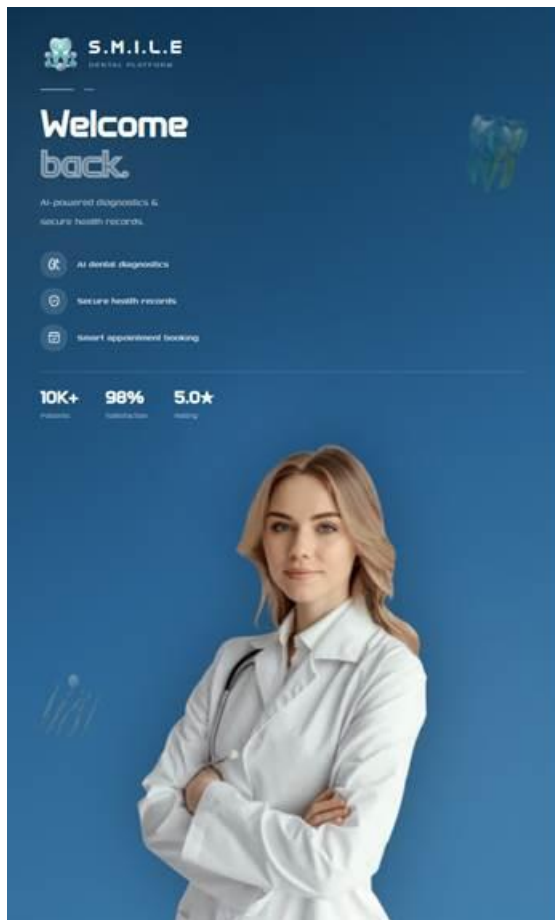


Figure 3.2-1: Staff login.

Step 2: Open Profile and select Identity Verification.

S.M.I.L.E.
SMILE PATIENT PLATFORM

Menu

- Dashboard
- Appointments
- Patients
- Imaging
- Examinations
- My Schedule

Search patients, files...

ACCOUNT
My Profile
Manage your personal information and settings

REDACTED
@mgp07hkgjcm00007733
ACTIVE DOCTOR

EMAIL verified PHONE verified

Profile info | Edit Profile | Change Password | Identity Verification

Identity Verification PENDING REVIEW

Verify your phone and submit citizen ID documents before booking.

SUBMITTED OCR CHECK ADMIN REVIEW PENDING REVIEW

Your documents were submitted successfully and need a manual review. Submission history

Phone Verification verified
Your phone number is verified.

CITIZEN ID FRONT
Verify front side of document
Capture

CITIZEN ID BACK
Verify back side of document
Capture

☐ I agree to the KVC document storage, OCR processing, forwarding utility, retention, and re-marketing terms.
[Read full terms](#)

Pending Review

Figure 3.2-2: Identity Verification form.

Step 3: Enter the identity document number, upload the front and back images, and provide consent.

The screenshot displays the 'My Profile' page of the S.M.I.L.E. Dental Platform. The page is divided into a left sidebar with navigation links (Menu, Dashboard, Appointments, Patients, Imaging, Examinations, My Schedule) and a main content area. The main content area features a search bar, a user profile card with 'ACTIVE' and 'VOCAL' status, and a navigation bar with links to 'Profile info', 'Edit Profile', 'Change Password', and 'Identity Verification'. The 'Identity Verification' section is highlighted and contains a progress bar with four steps: 'Document Upload', 'ID Document', 'Document Review', and 'Complete'. Below the progress bar, there is a section for 'Phone Verification' which is marked as 'Verified' with a green checkmark. The main form area has two large gray boxes for document capture, each with a 'Capture' button. At the bottom, there is a checkbox for 'I agree to the KYC document storage, OCR processing, biometric utility, retention, and de-sharing terms', a link to 'Read full terms', and a large blue 'Submit KYC' button. A 'NOT SUBMITTED' status is shown in the top right corner of the form area.

Figure 3.2-3: Completed verification form.

Step 4: Submit the verification request and confirm PENDING_REVIEW status.

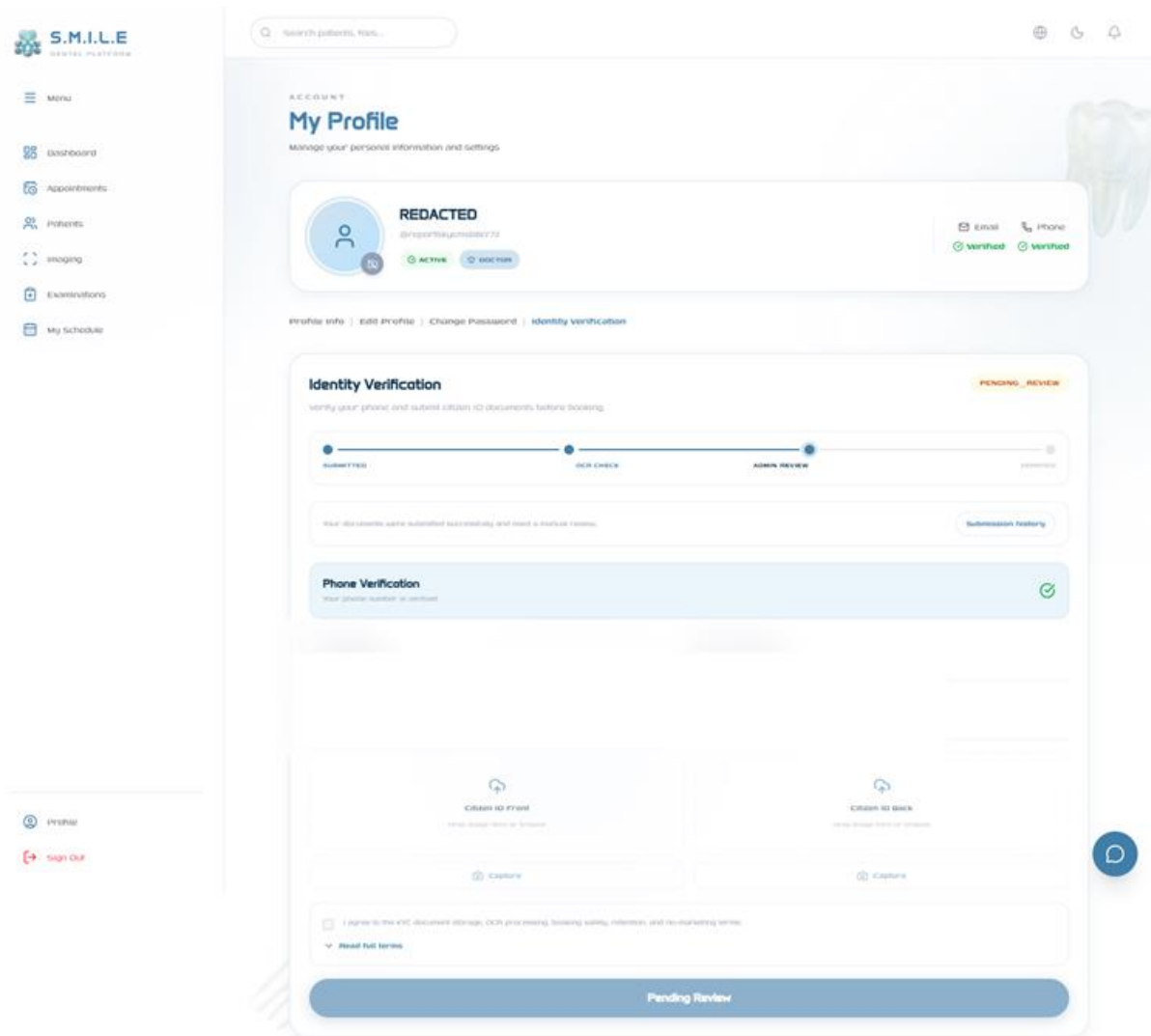


Figure 3.2-4: Submission pending review.

[Actor Switch: ADMIN]

Step 5: Open KYC Management and locate the pending submission.

Submitted name

REDACTED

Citizen ID

*****9012

OCR status

COMPLETED

Date of birth

REDACTED

Decision

Undecided

Confidence

—

OCR comparison

Document type

CITIZEN_ID

OCR full name

REDACTED

Risk

MEDIUM

OCR ID number

Detected

OCR date of birth

REDACTED

Decision reason

Manual review required: CONFIDENCE_UNAVAILABLE, RISK_NOT_LOW, NON_PASSING_CHECKS.

Automated checks

Front Side Present

Front image looks like the front side.

PASS

Back Side Present

Back image looks like the back side.

PASS

Front Back ID Match

Front ID number matches the ID2/back side ID number.

PASS

Submitted ID Match

Submitted ID number matches OCR result.

PASS

Submitted Dob Match

Submitted date of birth matches OCR result.

PASS

Front ID Number Found

Found a 12-digit Vietnamese citizen ID number.

Step 7: Approve the verified submission.

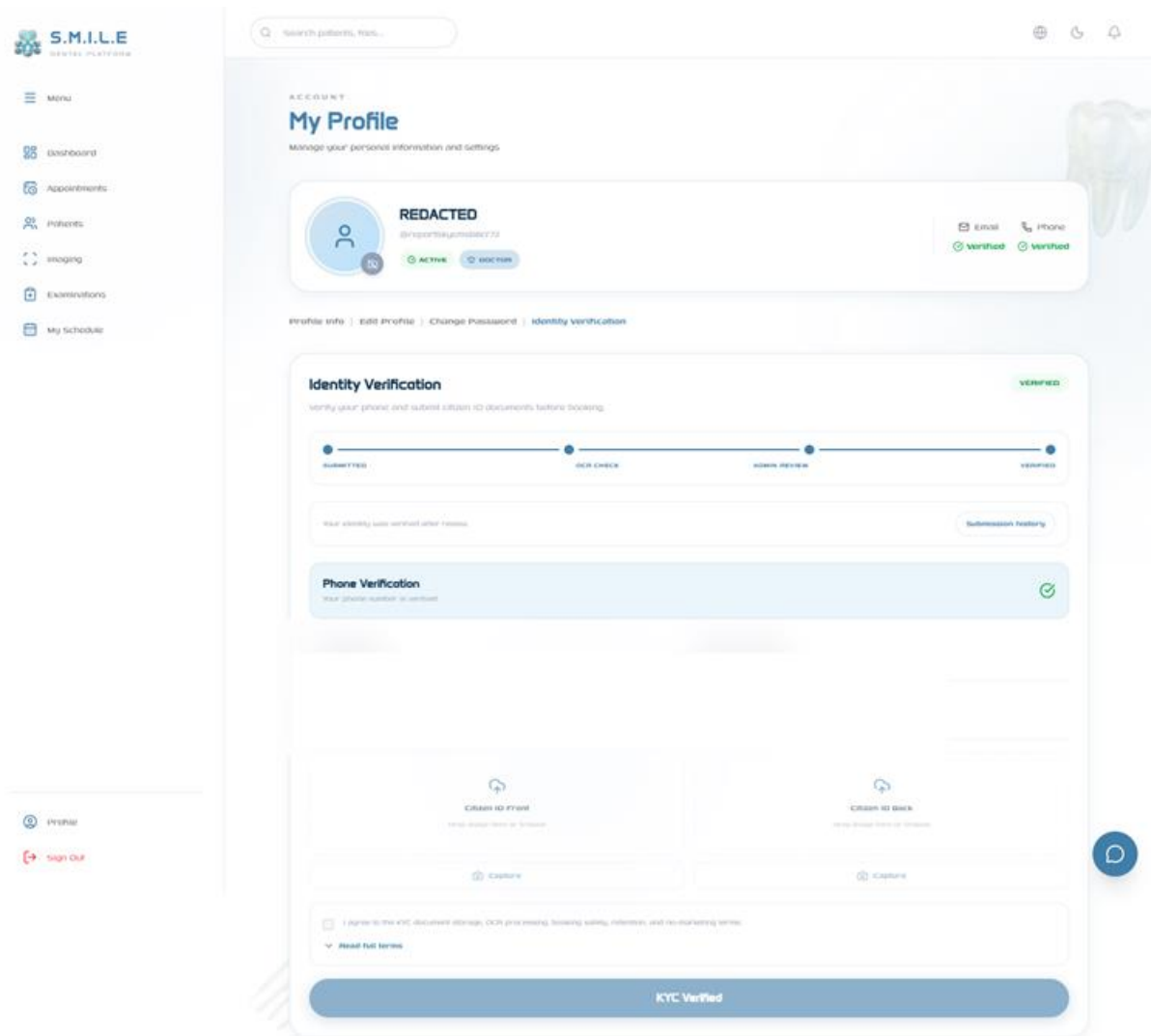


Figure 3.2-8: Verified staff profile.

3.3 FLOW 2 — APPOINTMENT BOOKING AND CONFIRMATION

3.3.1 Purpose and Outcome

This flow creates an appointment through the standard booking wizard and confirms the resulting schedule entry.

3.3.2 Step-by-Step Execution Guide

[Actor: PATIENT OR RECEPTIONIST]

Step 1: Open New Appointment and select a booking method.

The screenshot displays the S.M.I.L.E. 'Book an Appointment' interface. On the left is a sidebar menu with options: Dashboard, Appointments (highlighted), Patients, Imaging, Clinics, Specialists, Schedules, Examinations, Revenue, Performance, User Management, KYC Management, Refunds, Risk Management, and Audit Logs. At the bottom of the sidebar are 'Profile' and 'Sign Out' links. The main content area is titled 'Book an Appointment' with a subtitle: 'Follow the steps to choose how you'd like to schedule, fill the details, and confirm.' Below this is a progress bar with four steps: 1. Method, 2. Date, 3. Schedule, and 4. Patient & Review. Under the 'Method' step, there are four selection options: 'At Facility' (with a checkmark icon), 'By Specialty', 'By Doctor', and 'Outside Hours'. A 'Back' button is on the left and a 'Continue' button is on the right. At the bottom of the main area is a section titled 'Prefer chat? Try the Booking Assistant' with a brief description. A blue circular chat icon is located in the bottom right corner of the page.

Figure 3.3-1: Booking method selection.

Step 2: Select the clinic for the appointment.

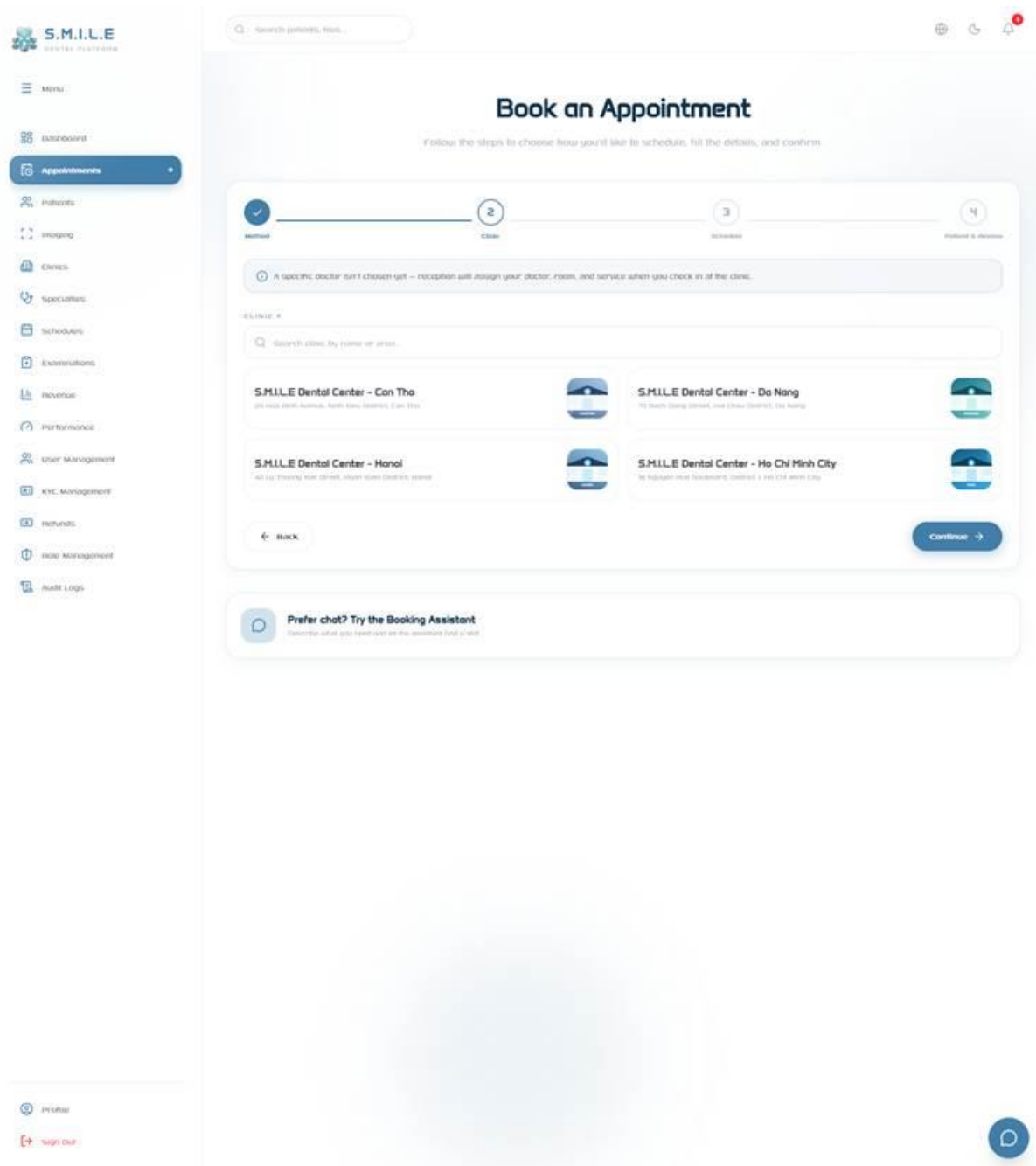


Figure 3.3-2: Clinic selection.

Step 3: Select the appointment date and available time slot.

The screenshot shows the S.M.I.L.E. 'Book an Appointment' interface. On the left is a sidebar menu with options: Menu, Dashboard, Appointments (highlighted), Patients, Imaging, Clinics, Specialists, Schedules, Examinations, Revenue, Performance, User Management, KYC Management, Refunds, Risk Management, and Audit Logs. At the bottom of the sidebar are 'Profile' and 'Sign Out' links. The main content area has a search bar at the top and a progress bar with four steps: Method, Clinic, Schedule (active), and Patient & Review. The 'Schedule' step contains a note: 'A specific doctor isn't chosen yet - reception will assign your doctor, room, and service when you check in at the clinic.' Below this are fields for 'SERVICE (OPTIONAL)' (set to 'No specific service'), 'DATE' (set to 'Thu, 06 Aug 2026'), and 'TIME' (set to '10:00'). 'Back' and 'Continue' buttons are at the bottom of the form. A 'Prefer chat? Try the Booking Assistant' link is also present. A floating chat button is in the bottom right corner.

S.M.I.L.E.
Smile Platform

Menu

Dashboard

Appointments

Patients

Imaging

Clinics

Specialists

Schedules

Examinations

Revenue

Performance

User Management

KYC Management

Refunds

Risk Management

Audit Logs

Profile

Sign Out

Search patients, files...

Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm

Method Clinic **Schedule** Patient & Review

A specific doctor isn't chosen yet - reception will assign your doctor, room, and service when you check in at the clinic.

SERVICE (OPTIONAL)
No specific service

DATE
Thu, 06 Aug 2026

TIME
10:00

Back Continue

Prefer chat? Try the Booking Assistant
Discover what you need and get the answers fast & easy

Figure 3.3-3: Date and time selection.

Step 4: Select the patient and verify the booking summary.



Menu

Dashboard

Appointments

Patients

Imaging

Clinics

Specialties

Schedules

Examinations

Revenue

Performance

User Management

KYC Management

Refunds

Risk Management

Audit Logs

Search patients, files...

Book an Appointment

Follow the steps to choose how you'd like to schedule, fill the details, and confirm.

Method

Class

Schedule

Patient & Review

1

A specific doctor isn't chosen yet – reception will assign your doctor, room, and service when you check in at the clinic.

PATIENT &

NOTE

Search patient by name or code...

Additional notes

Lucas Nguyen (VIN-2526-0000)

REVIEW

Method

At Facility

Room

Lucas Nguyen

CLINIC

S.M.I.L.E Dental Center - Can Tho

DATE

2025-09-06

Time

10:00

BACK

Confirm Booking

2

Prefer chat? Try the Booking Assistant

Describe what you need and let the assistant help you.

Profile

Sign Out

Figure 3.3-4: Patient and booking review.

Step 5: Submit the booking and confirm the new appointment in the list.

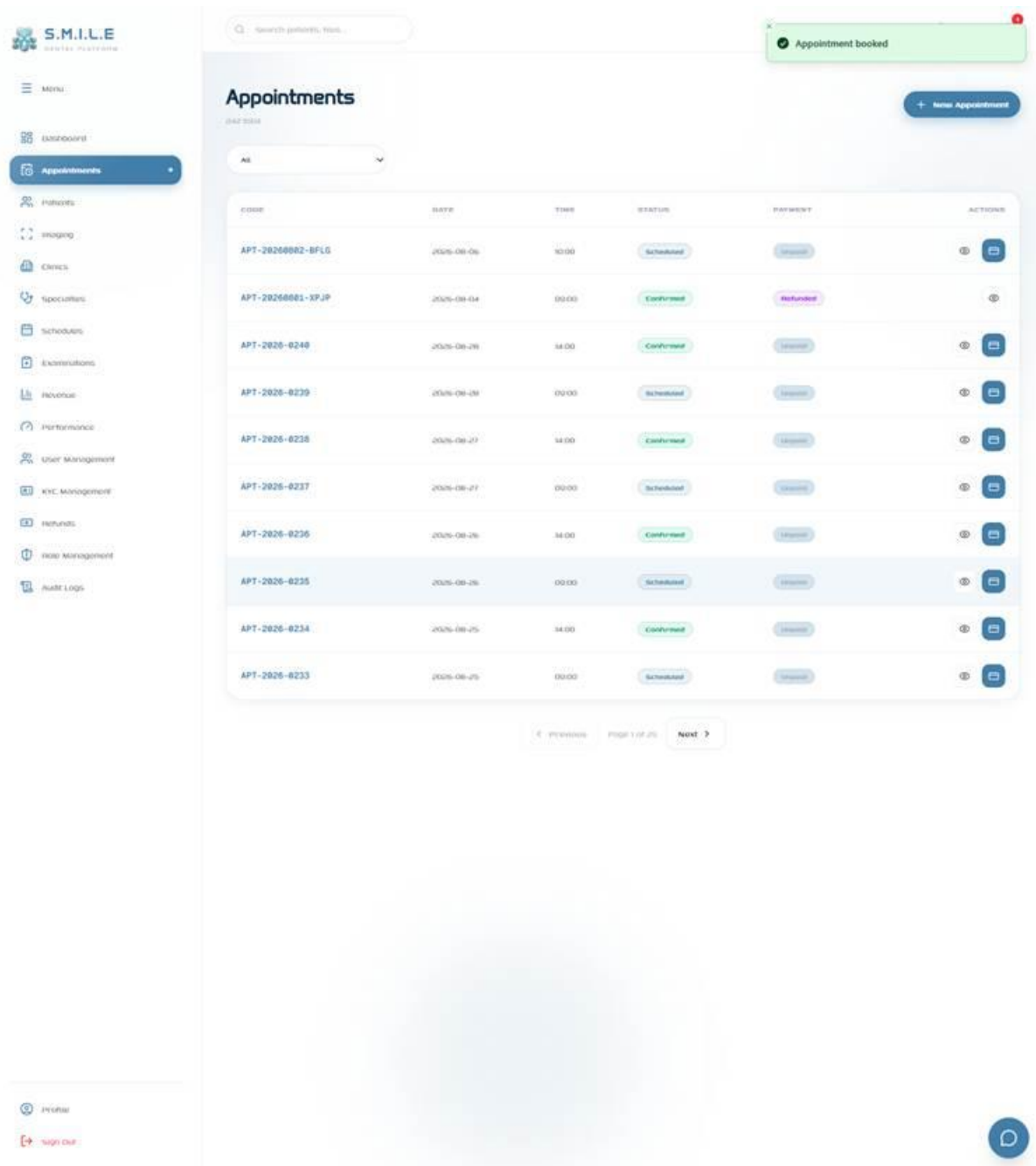


Figure 3.3-5: Appointment booking result.

3.4 FLOW 3 — CLINICAL VISIT: FROM CHECK-IN TO FINALIZED RECORD

3.4.1 Purpose and Outcome

This flow records patient arrival, clinical findings, diagnosis, treatment planning, and the finalized examination record.

3.4.2 Step-by-Step Execution Guide

[Actor: RECEPTIONIST]

Step 1: Open the appointment and verify its schedule and payment status.

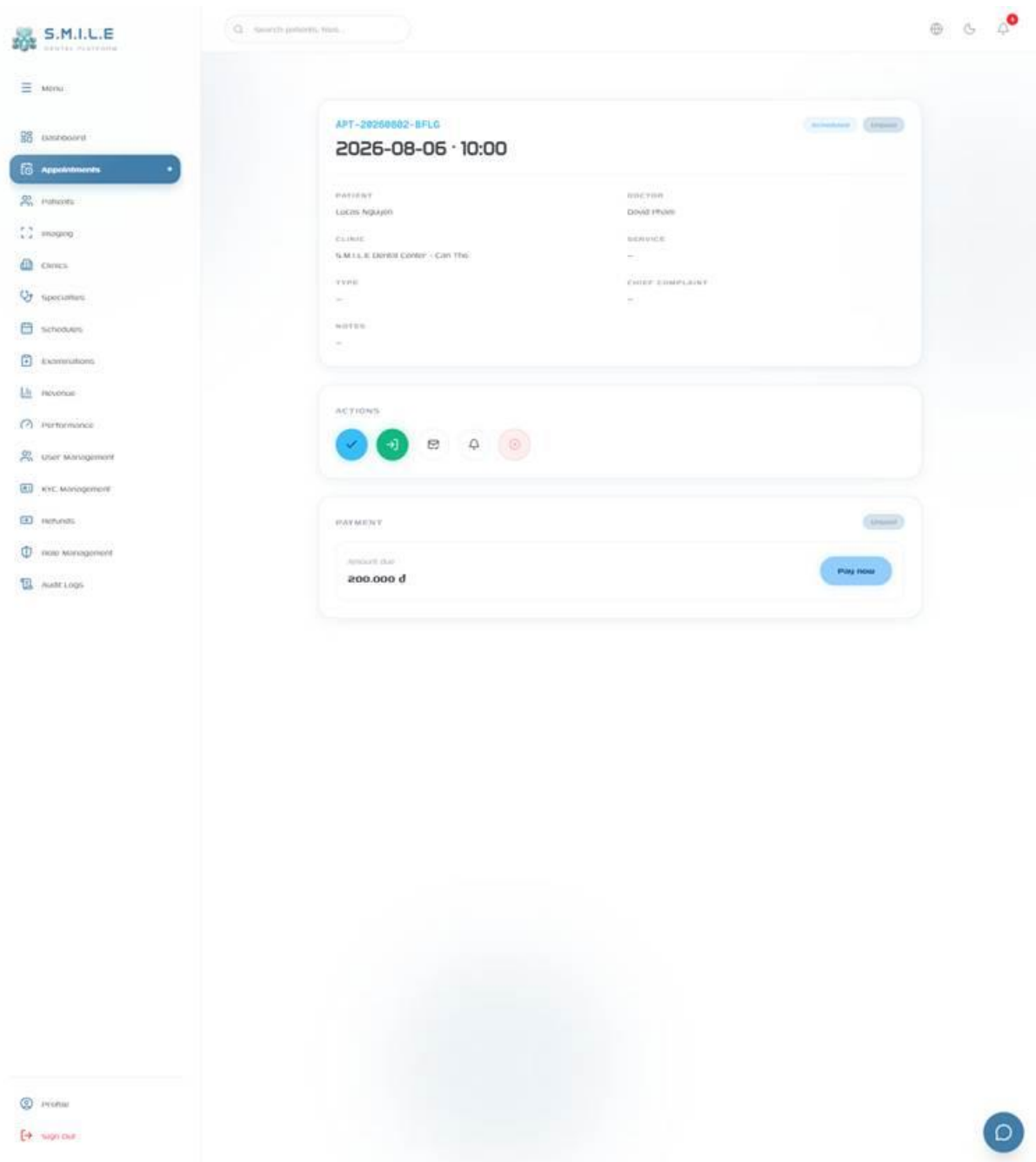


Figure 3.4-1: Scheduled appointment details.

Step 2: Open Assign & Check In and confirm the assigned doctor, service, and room.

The screenshot displays the S.M.I.L.E. Patient check-in form. The interface includes a sidebar menu on the left with options like Dashboard, Appointments, Patients, Imaging, Clinics, Specialists, Schedules, Examinations, Revenue, Performance, User Management, KYC Management, Refunds, Risk Management, and Audit Logs. The main content area shows the appointment details for APT-20250802-BFLG on 2026-08-06 at 10:00. The patient is LOCUS NGUYEN, the doctor is DVID PHAM, and the clinic is S.M.I.L.E ERWIN CORP - Can Tho. The form includes sections for ACTIONS (with icons for check, message, alert, and settings), ASSIGN & CHECK IN (with dropdowns for Doctor, Service, and Treatment Room), and PAYMENT (showing an amount due of 200,000 d and a Pay Now button). The bottom of the screen features a PREVIEW button and a Logout button.

S.M.I.L.E.
Smile Platform

Menu

- Dashboard
- Appointments**
- Patients
- Imaging
- Clinics
- Specialists
- Schedules
- Examinations
- Revenue
- Performance
- User Management
- KYC Management
- Refunds
- Risk Management
- Audit Logs

Search patients, New...

APT-20250802-BFLG

2026-08-06 · 10:00

PATIENT
LOCUS NGUYEN

DOCTOR
DVID PHAM

CLINIC
S.M.I.L.E ERWIN CORP - Can Tho

SERVICE

TYPE

CHIEF COMPLAINT

NOTES

ACTIONS

✓ ⓘ 🔔 ⚙️

ASSIGN & CHECK IN

Choose which doctor and service the patient will use today before checking them in.

DOCTOR *
Dvid Pham

SERVICE (OPTIONAL)
No specific service

TREATMENT ROOM (OPTIONAL)
Auto-assign from doctor's schedule

Continue Check in Cancel

PAYMENT

Amount due:
200,000 d

Pay now

PREVIEW

Logout

Figure 3.4-2: Patient check-in form.

Step 3: Confirm that the appointment status changes to Checked In.

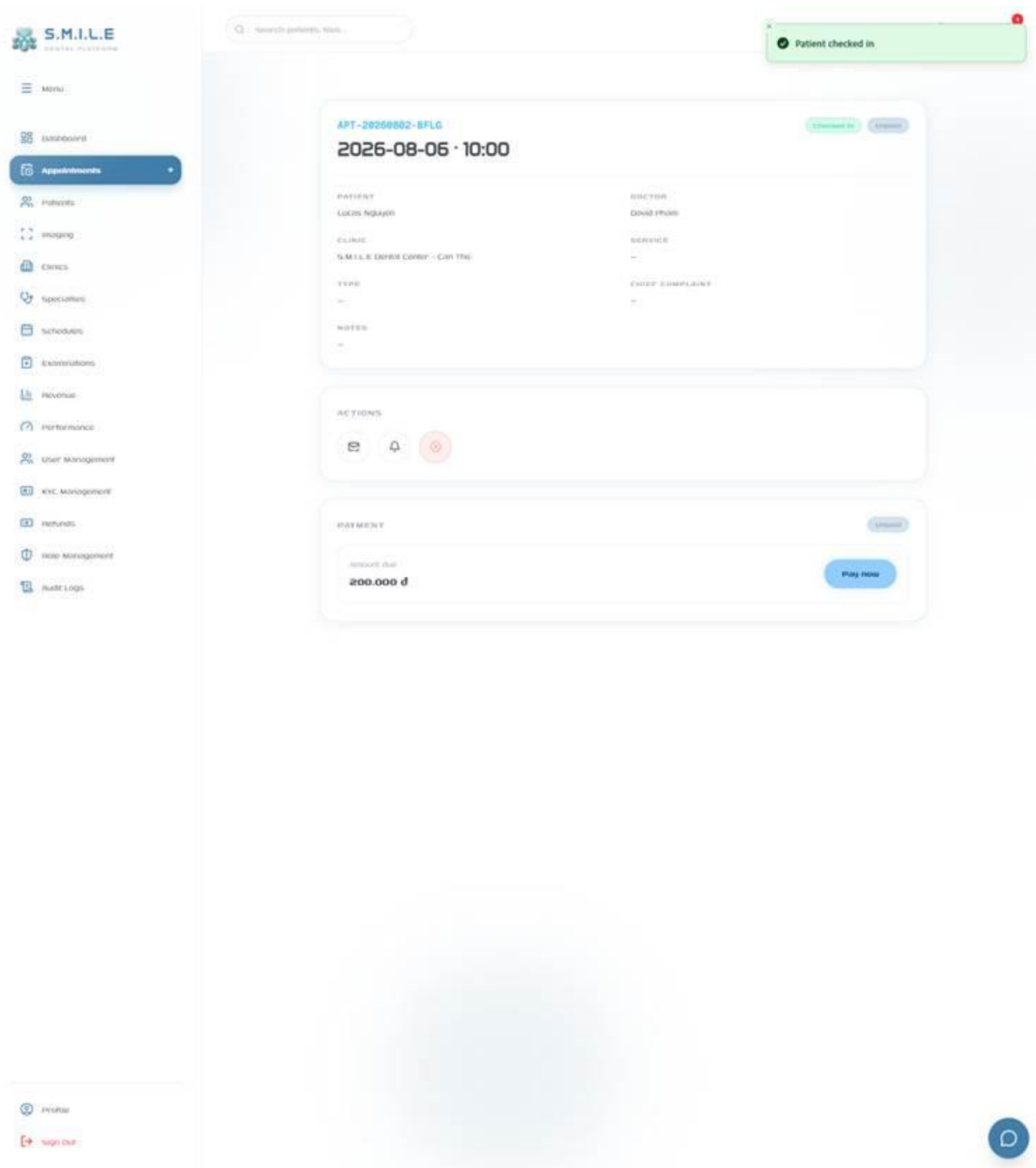


Figure 3.4-3: Checked-in appointment.

[Actor Switch: DOCTOR]

Step 4: Open the clinical examination and review the clinical notes, alerts, and representative information.

S.M.I.L.E.
Smile Platform

Search patients, files...

← BACK TO EXAMINATIONS

Clinical Examination

72x60455 Completed

Luke Nguyen Doctor (Dentist) 10/10/2024

Chief complaint: Intermittent sensitivity and pain when chewing on the upper right molar.

[Add amendment](#) [Schedule recall](#) [Print report](#)

Clinical notes

CHIEF COMPLAINT
Intermittent sensitivity and pain when chewing on the upper right molar.

PRESENT ILLNESS
Sensitivity began two weeks ago after a fractured restoration. Pain increases with cold drinks and chewing.

PHYSICAL EXAMINATION
Upper right first molar has a fractured occlusal restoration. Tooth is vital with mild percussion tenderness. Gingiva is healthy.

[Save notes](#)

Clinical alerts

ALLERGIES
Penicillin

CHRONIC DISEASES
Controlled Hypertension

MEDICAL HISTORY
Routine dental care history (medical history reviewed during the dental intake process).

[Review before treatment](#)

Legal representative

Ava Nguyen Primary Verified

[Add](#)

Recall reminders

Reminder enabled ☒ 1 Day

[Save preference](#) [Send now](#) [History failed](#)

No recall reminder activity recorded.

Amendments [0]

No amendments recorded for this encounter.

[Add amendment](#)

Follow-up / Recall [0]

No follow-up appointment linked to this encounter.

[Schedule recall](#)

Symptoms [0]

[Enter symptom](#)

Profile [Sign out](#)

Figure 3.4-4: Clinical examination record.

Step 5: Review symptoms, diagnosis, treatment plan, prescription, dental chart, and diagnostic orders before closing the record.

The screenshot displays the S.M.I.L.E. (Smart Medical Information Linkage Engine) interface. On the left is a sidebar menu with options: Home, Dashboard, Appointments, Patients, Imaging, Clinics, Specialists, Schedulers, **Examinations** (highlighted), Revenue, Performance, User Management, KYC Management, Refunds, Risk Management, and Audit Logs. At the bottom of the sidebar are buttons for 'Profile' and 'Sign Out'.

The main content area shows a search bar at the top with the text 'Search patients, files...'. Below it, several sections are visible, each with a title, a list of items, and an action button:

- Symptoms (1)**: Contains one entry, 'Tooth sensitivity', with details: 'Upper right first molar', '2 weeks', '2/16/2020', and 'Triggered by cold drinks and chewing'. Action button: '+ Enter symptom'.
- Diagnoses (1)**: Contains one entry, 'Cracked upper right first molar', with details: 'M1.21', 'Primary', and 'Cracks extend through the chewing surface along occlusal cuspal embrasure'. Action button: '+ Add diagnosis'.
- Treatment Plans (1)**: Contains one entry, 'Ceramic crown restoration', with details: 'Permanent', '2 weeks', '\$1,000.00 (I - on payment 50%/50%)', and 'Assigned to 2/16/2020'. Description: 'Protect the cracked tooth, restore chewing function, and prevent further fracture. Note: (crown) state of: none documented, alternatives documented'. Action button: '+ Create plan'.
- Prescription (1)**: Contains one entry, 'Paracetamol', with details: '1000 mg, 3 times a day, as needed', 'oral', '3 days', '2/16/20', and 'Take after meals only when pain is present. Do not exceed 3000 mg per day'. Action button: '+ Create electronic prescription'. There is also a '+ Add drug' button.
- Dental Chart (1)**: Contains one entry, 'Tooth 16', with details: 'Fixed' and 'Fractured occlusal restoration with a crack across the chewing surface'. Action button: '+ Chart tooth'.
- Diagnostic Orders — X-ray / CBCT (1)**: Contains one entry, 'X RAY - DO-XRAY-2026001-9TCR', with details: 'Permanent', 'Tooth 16', 'Upper right first molar', and 'Periapical radiograph'. Action button: '+ Order X-ray / CBCT'.
- Clinical / Lab Orders (0)**: Contains no entries. Action buttons: '+ Order Lab Test' and '+ Order Clinical Test'.

At the bottom right of the main area is a circular button with a magnifying glass icon.

Figure 3.4-5: Finalized diagnosis and treatment record.

3.5 FLOW 4 — PAYMENT AND REFUND

3.5.1 Purpose and Outcome

This flow verifies the appointment payment state and records an administrator refund decision.

3.5.2 Step-by-Step Execution Guide

[Actor: PATIENT OR RECEPTIONIST]

Step 1: Open Payment and verify the appointment summary and available payment method.



Menu

Dashboard

Appointments

Patients

Imaging

Clinics

Specialties

Schedules

Examinations

Revenue

Performance

User Management

KYC Management

Refunds

Risk Management

Audit Logs

Search patients, files...

Back

APT-20260802-BFLG

Payment

Verify your appointment and continue through the secure payment gateway.

PAYMENT SUMMARY

Appointment Code

APT-20260802-BFLG

Date / Time

2026-08-06 10:00

Service

Not yet specified

Doctor

David Pham

Clinic

S.M.I.L.E Dental Center - Can Tho

PAYMENT METHOD

VnPay

Pay by domestic card, international card, or VnPay QR

Momo

Coming soon

ZaloPay

Coming soon

Total Amount

Secure Payment

VnPay processes the encrypted transaction. S.M.I.L.E does not store your card information.

Continue to VnPay

Cancel

Profile

Sign Out

Figure 3.5-1: Payment summary and method.

Step 2: Confirm the completed payment state before further processing.

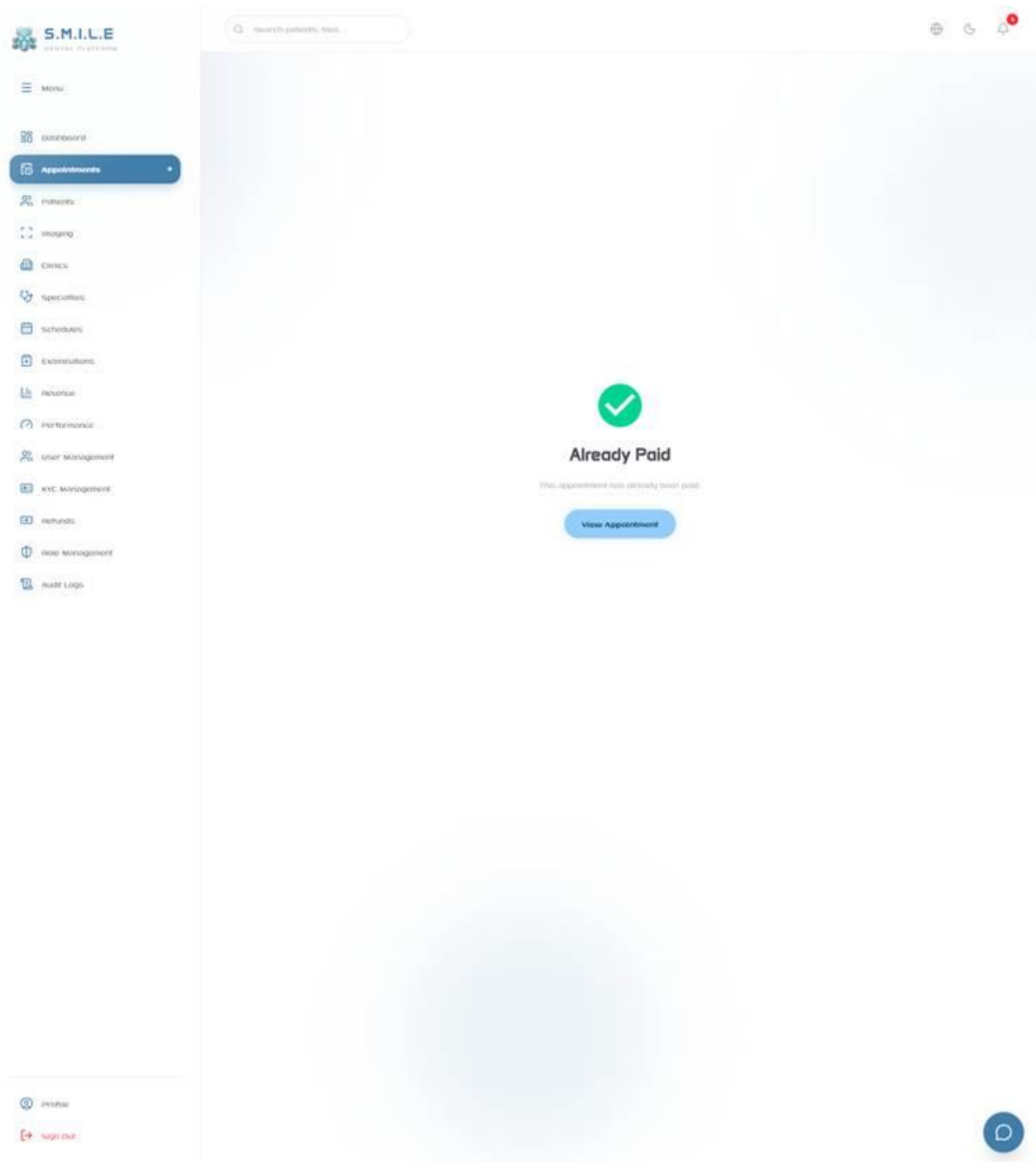


Figure 3.5-2: Completed payment status.

[Actor Switch: ADMIN]

Step 3: Open Refunds and review pending requests, amounts, reasons, and available decisions.

- Menu
- Dashboard
- Appointments
- Patients
- Imaging
- Clinics
- Specialties
- Schedulers
- Examinations
- Revenue
- Performance
- User Management
- KYC Management
- Refunds**
- Risk Management
- Audit Logs

Refund approvals

11 requests pending review

11

Pending

Under Review

Approved

Refunding

Refunded

Rejected

PAYMENT ID	AMOUNT	REASON	REQUESTED AT	STATUS	ACTIONS
e3d6a72...	8,000,000 ₪	Crown treatment was postponed after the fill...	Aug 01, 2026, 02:07 PM	Refunded	
25cd7b43...	200,000 ₪	The appointment scheduler changed after pay...	Aug 01, 2026, 04:40 PM	Pending	✓ Approve ✗ Reject
e5a85fab...	200,000 ₪	Rescheduled	Aug 01, 2026, 04:50 AM	Pending	✓ Approve ✗ Reject
a0000000...	450,000 ₪	Patient requested a review of the final treatm...	Jul 05, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	300,000 ₪	Appointment cancelled before treatment was ...	Jul 24, 2026, 04:00 PM	Refunded	
a0000000...	3,200,000 ₪	Patient requested a review of the final treatm...	Jul 05, 2026, 03:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	8,000,000 ₪	Appointment cancelled before treatment was ...	Jul 01, 2026, 04:00 PM	Refunded	
a0000000...	40,000,000 ₪	Patient requested a review of the final treatm...	Jul 06, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	10,000,000 ₪	Patient requested a review of the final treatm...	Jul 26, 2026, 03:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	4,000,000 ₪	Appointment cancelled before treatment was ...	Jul 03, 2026, 03:00 PM	Refunded	
a0000000...	500,000 ₪	Patient requested a review of the final treatm...	Jul 17, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	900,000 ₪	Patient requested a review of the final treatm...	Jul 07, 2026, 02:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	350,000 ₪	Appointment cancelled before treatment was ...	Jul 03, 2026, 03:00 PM	Refunded	
a0000000...	800,000 ₪	Patient requested a review of the final treatm...	May 26, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	55,000,000 ₪	Patient requested a review of the final treatm...	May 03, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	3,000,000 ₪	Appointment cancelled before treatment was ...	May 17, 2026, 03:00 PM	Refunded	
a0000000...	200,000 ₪	Patient requested a review of the final treatm...	May 31, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject

Profile

Sign Out

Figure 3.5-3: Refund approval queue.

Step 4: Approve the valid request and confirm that its status changes to Refunded.

PAYMENT ID	AMOUNT	REASON	REQUESTED AT	STATUS	ACTIONS
c3d0472...	8,000,000 ₦	Crown treatment was postponed after the fill...	Aug 01, 2026, 02:07 AM	Refunded	
75cd7b43...	200,000 ₦	The appointment schedule changed after pay...	Aug 01, 2026, 04:40 PM	Refunded	
e5a85fab...	200,000 ₦	requested	Aug 01, 2026, 10:10 AM	Pending	✓ Approve ✗ Reject
a0000000...	450,000 ₦	patient requested a review of the final treatm...	Jul 29, 2026, 11:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	300,000 ₦	Appointment cancelled before treatment was ...	Jul 29, 2026, 03:00 PM	Refunded	
a0000000...	3,200,000 ₦	Patient requested a review of the final treatm...	Jul 29, 2026, 03:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	8,000,000 ₦	Appointment cancelled before treatment was ...	Jul 07, 2026, 03:00 PM	Refunded	
a0000000...	40,000,000 ₦	Patient requested a review of the final treatm...	Jul 06, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	12,000,000 ₦	patient requested a review of the final treatm...	Jun 26, 2026, 09:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	4,000,000 ₦	Appointment cancelled before treatment was ...	Jun 25, 2026, 09:00 PM	Refunded	
a0000000...	500,000 ₦	patient requested a review of the final treatm...	Jun 17, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	900,000 ₦	patient requested a review of the final treatm...	Jun 07, 2026, 02:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	350,000 ₦	Appointment cancelled before treatment was ...	Jun 03, 2026, 09:00 PM	Refunded	
a0000000...	800,000 ₦	patient requested a review of the final treatm...	May 29, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	55,000,000 ₦	Patient requested a review of the final treatm...	May 19, 2026, 09:00 PM	Pending	✓ Approve ✗ Reject
a0000000...	3,000,000 ₦	Appointment cancelled before treatment was ...	May 17, 2026, 03:00 PM	Refunded	
a0000000...	200,000 ₦	patient requested a review of the final treatm...	May 15, 2026, 04:00 PM	Pending	✓ Approve ✗ Reject

Figure 3.5-4: Completed refund status.

3.6 FLOW 5 — CLINIC ADMINISTRATION AND OVERSIGHT

3.6.1 Purpose and Outcome

This flow covers the principal configuration, access-control, financial, performance, and audit functions available to an administrator.

3.6.2 Step-by-Step Execution Guide

[Actor: ADMIN]

Step 1: Open Dashboard and review the current operational and revenue summary.

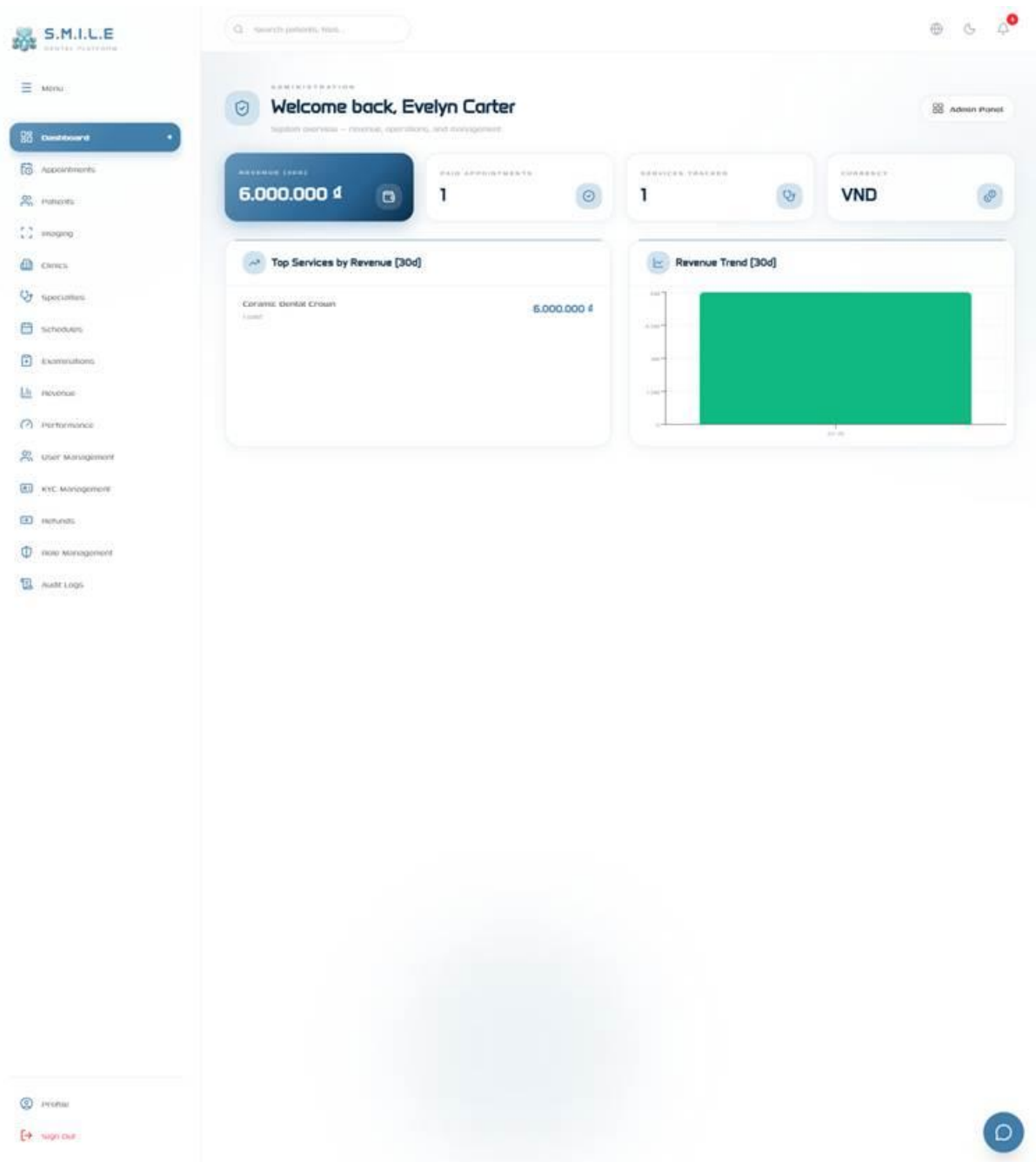


Figure 3.6-1: Administration dashboard.

Step 2: Open Clinics to review active locations and clinic details.

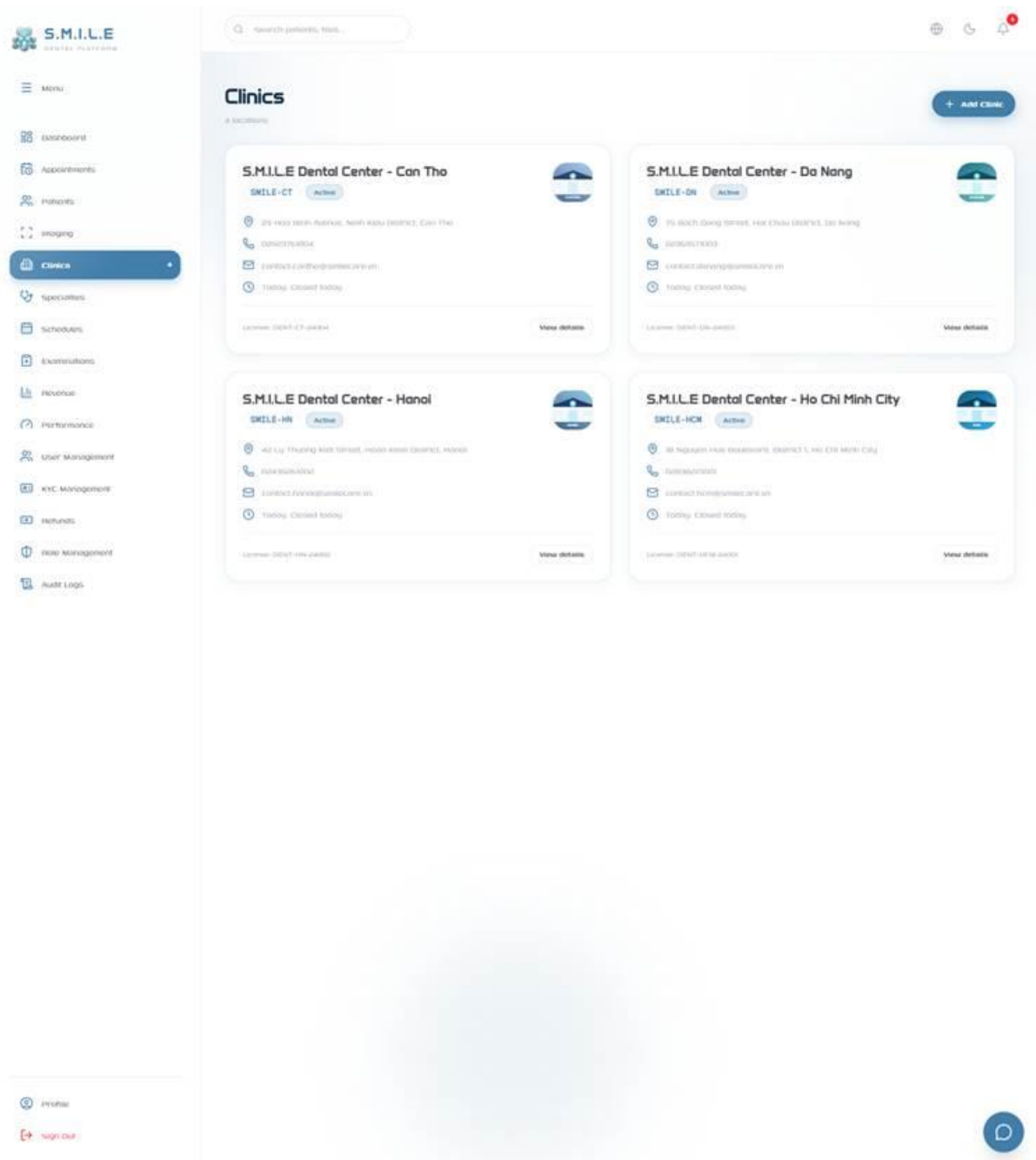


Figure 3.6-2: Clinic management.

Step 3: Open Specialties to review available clinical specialties.

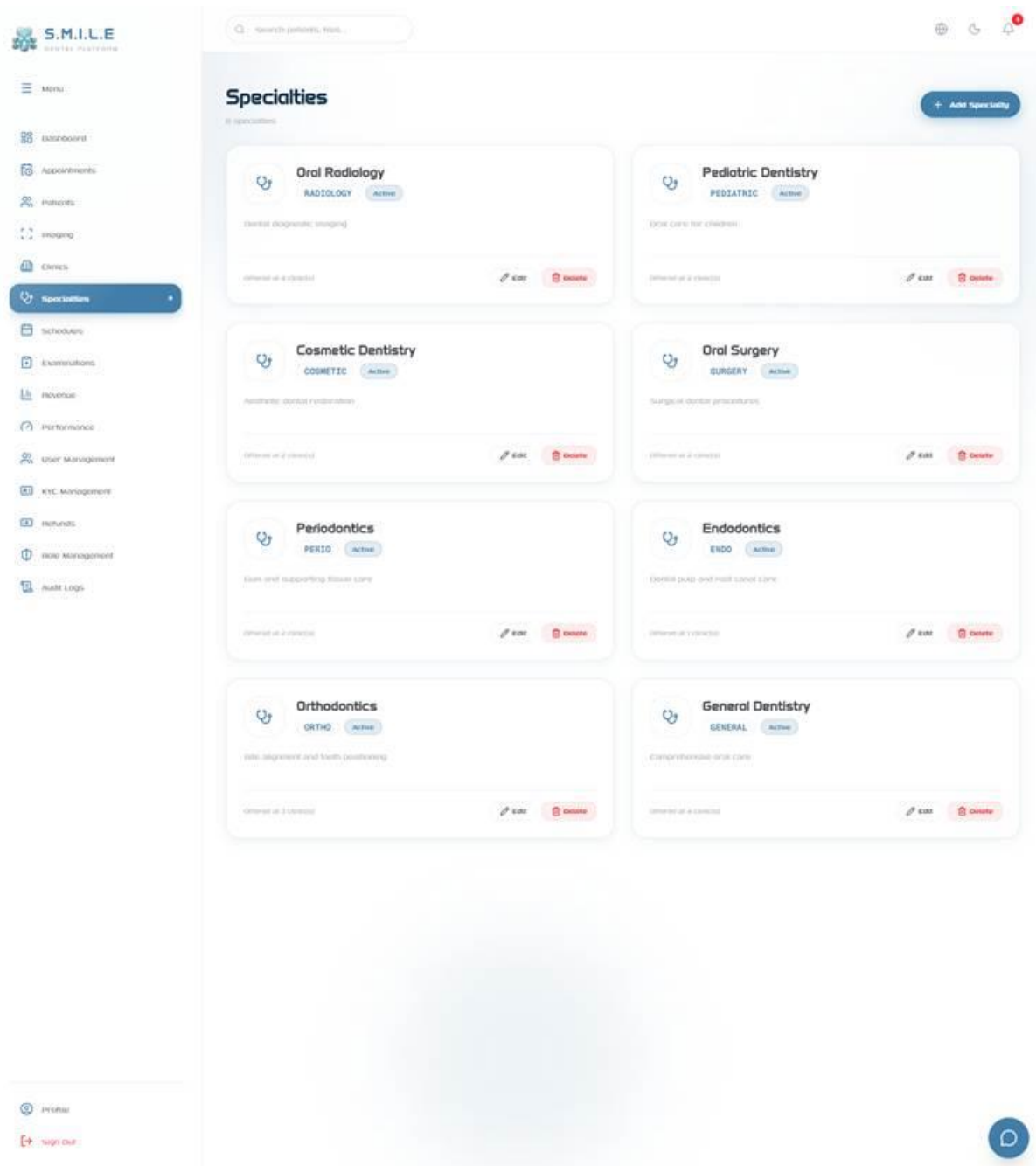


Figure 3.6-3: Specialty management.

Step 4: Open Schedules to access work schedules, shifts, and leave management.

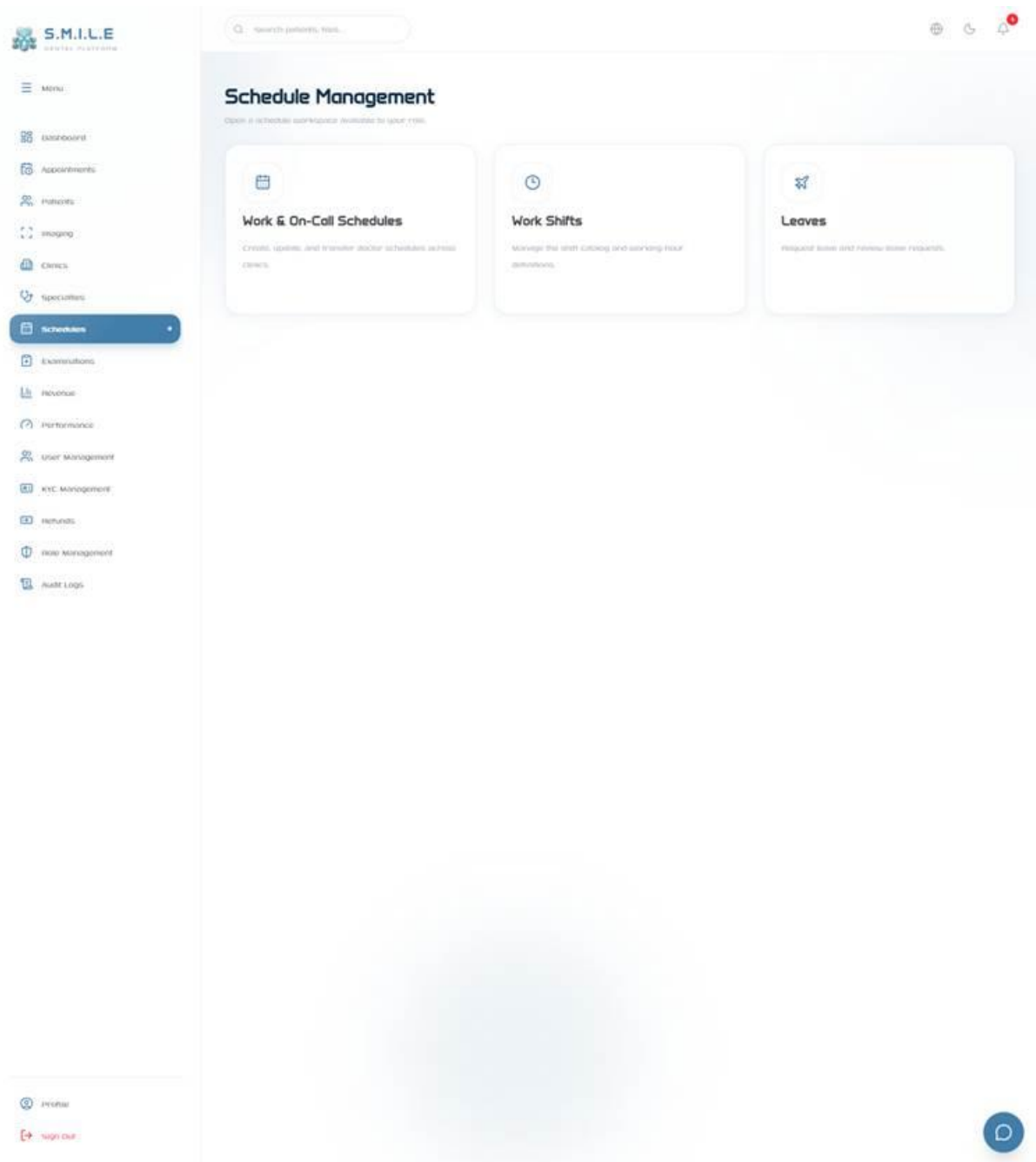


Figure 3.6-4: Schedule management.

Step 5: Open User Management to search accounts and manage access status.

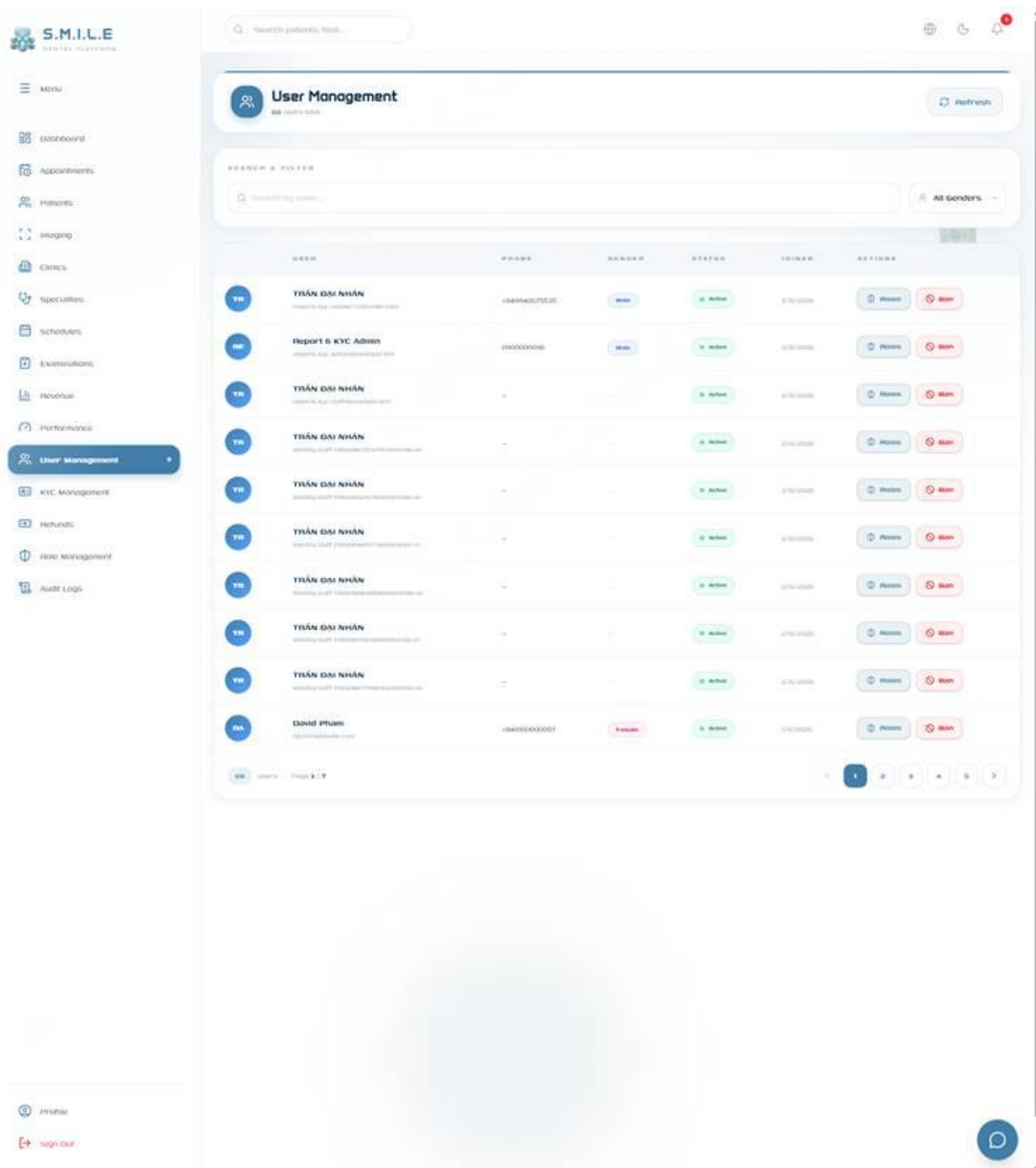


Figure 3.6-5: User management.

Step 6: Open Role Management to review roles and permission controls.

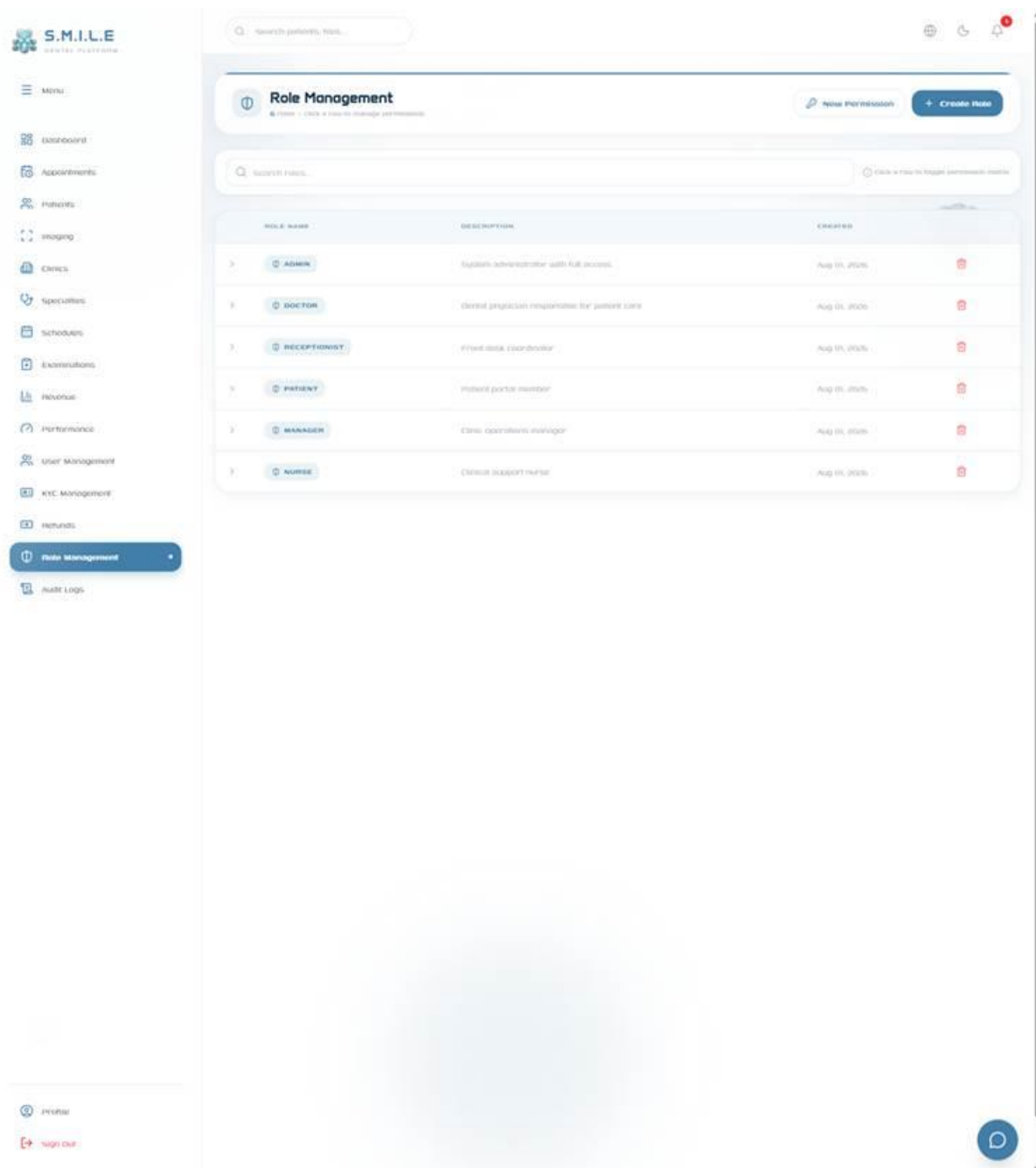


Figure 3.6-6: Role management.

Step 7: Open Revenue Reports to review paid appointments and revenue by service and clinic.

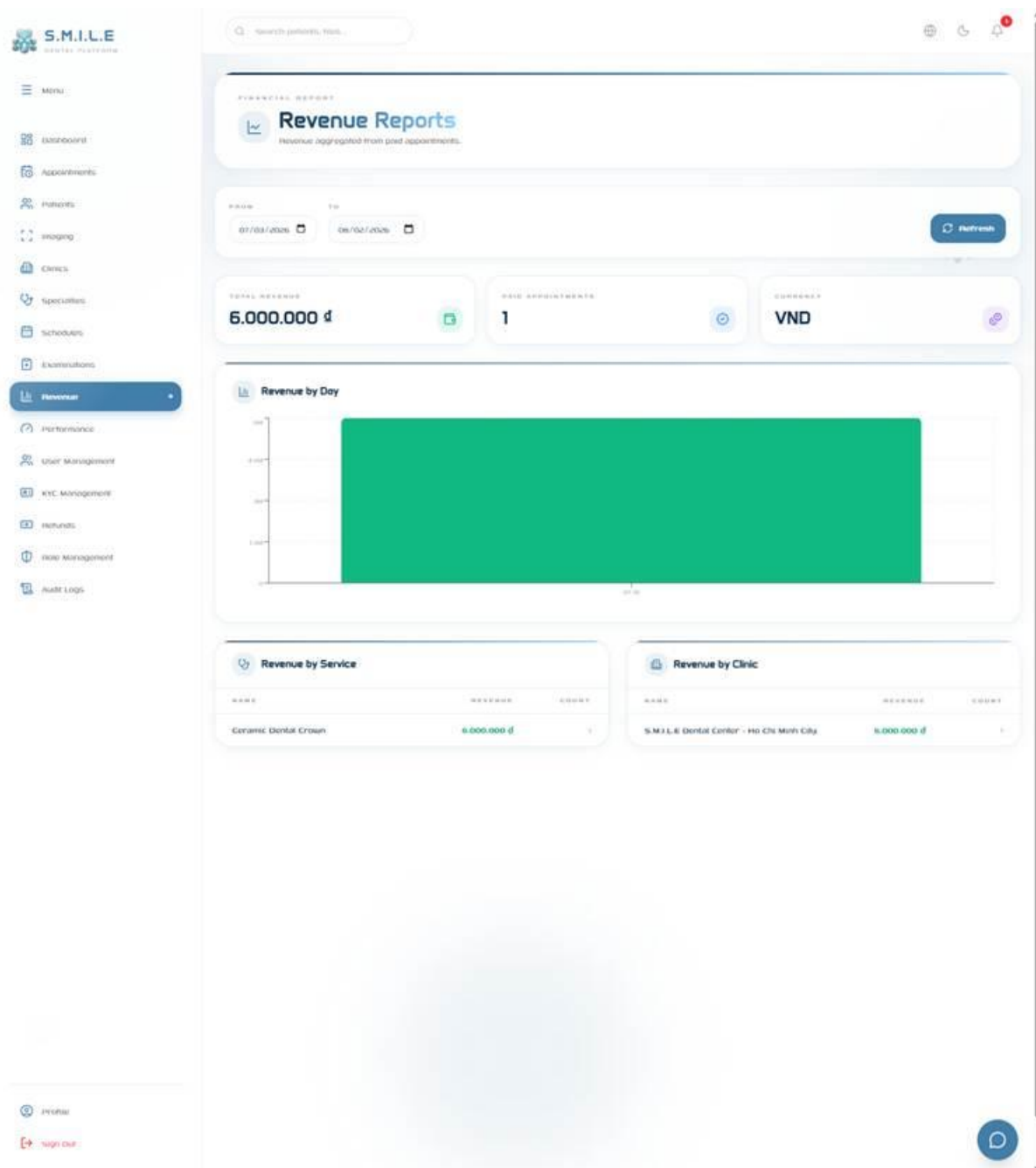


Figure 3.6-7: Revenue report.

Step 8: Open Performance to review completion and cancellation indicators.

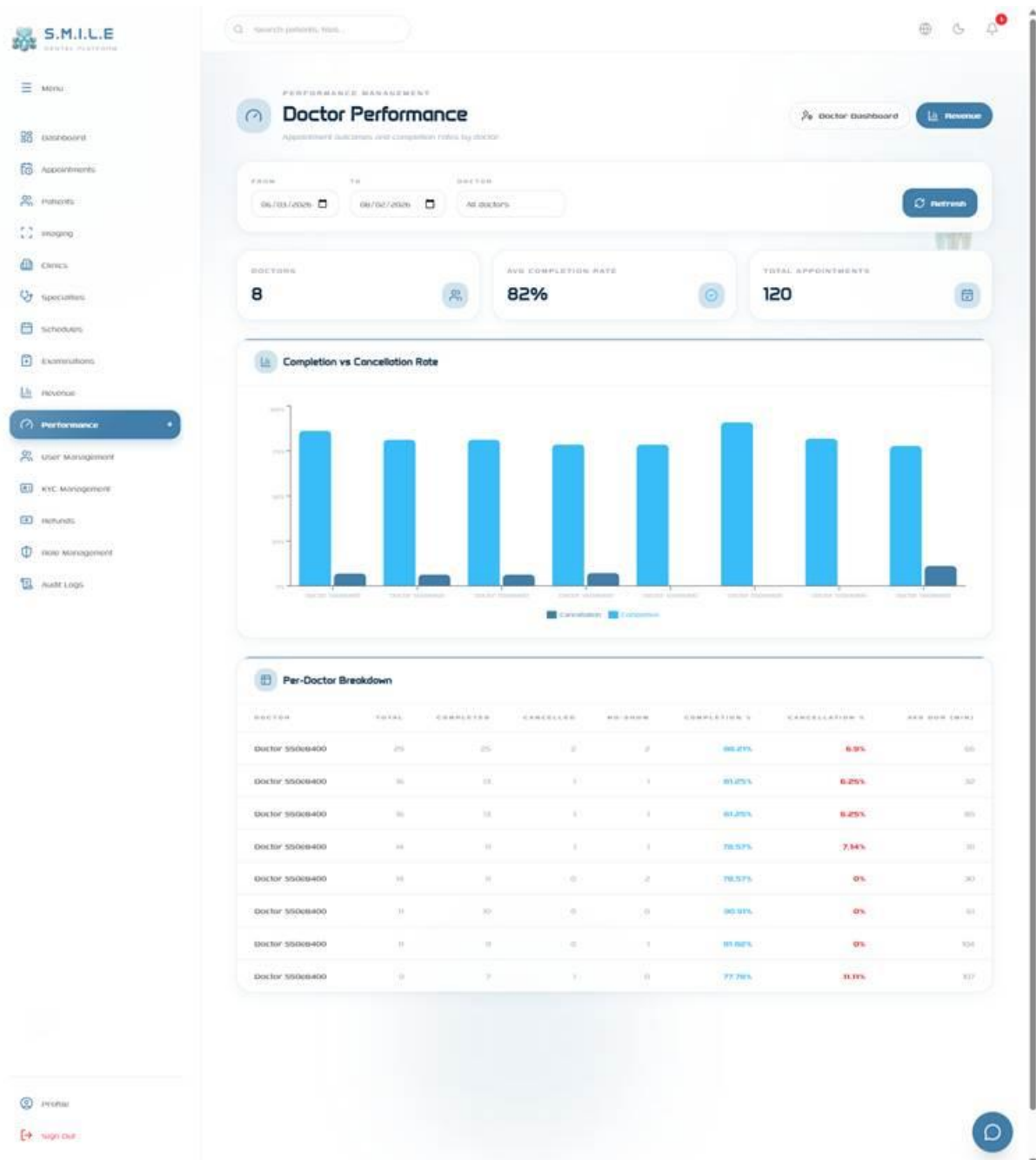


Figure 3.6-8: Doctor performance report.

Step 9: Open Audit Logs to review recorded access and administrative actions.

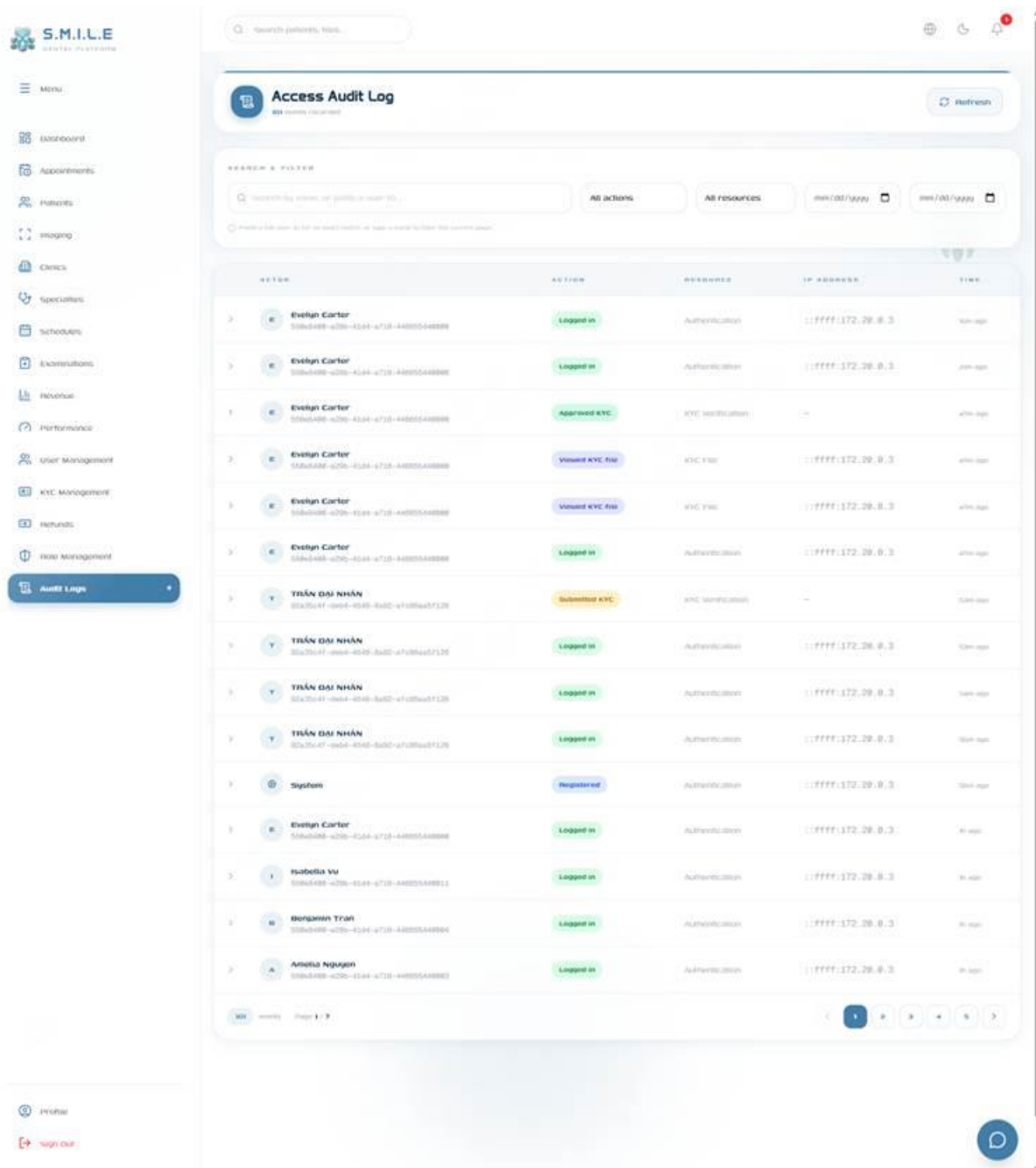


Figure 3.6-9: Access audit log.

3.7 CURRENT LIMITATIONS AND TEST BOUNDARIES